# Week 1

1. Make DAG
    a. Response variables (E.G, Egg masses)
    b. Covariates (E.G, Rainfall)
2. Site Conditions (Covariates)
    a. L
3. GIS and Cost Surface Model
    a. Low Cost
    b. High Cost
4. Extinction Probability
    a. MARSS
        i. Quasi-extinction probability

# Week 2

June 8th, 2026

1. Anthony's Proposal
    a. Phase 1: Local Vulnerability (Covariate Matrix)
        i. (Site specific Environmental Stress elements)
            1. Current Year CWD
                a. Speed of water loss
                b. Darren - look at slope of staff plate for when gets to zero?
                c. Staff plates for proxies for other nearby sites
                d. Interpolation or extrapolation for missing data
            2. 3 year lagged rain
                a. Historical input of female breeding cohort
                b. Egg→ Metamorpho around 6 months
            3. Hydroperiod (hab type) seasonal or permanent wetland
                a. Whether or not site has a water buffer
    b. Phase 2: Spatial Connectivity
        i. (Dispersal/Transition Matrix)
        ii. 10-m Resolution GIS Surface
            1. Veg Cover
            2. Slope
            3. Wet season travel vectors
                a. NHD Stream Vectors
                b. Wetlands potential corridor
                    i. 2018 data package

ii. Veg classes associated with wetlands

1. Weight higher than other drier veg types

4. Max 2,800m min limit (Fellers telemetry Data)
5. Can travel point A to point B due to fogginess of coastal environment?
6. Use Genomes to ground truth and calibrate the paths

c. Phase 3: Integrations

i. Feed time series 1997-2025 into MARSS

1. Train model on 1997-2020 data

a. Validate predictive accuracy off of 2021-2025

2. Observation Error vs Process Variance (Biology)
3. Would pick up the Newts eating the egg masses even if not explicitly modeled in

d. Phase 4: Scenario Forecasting

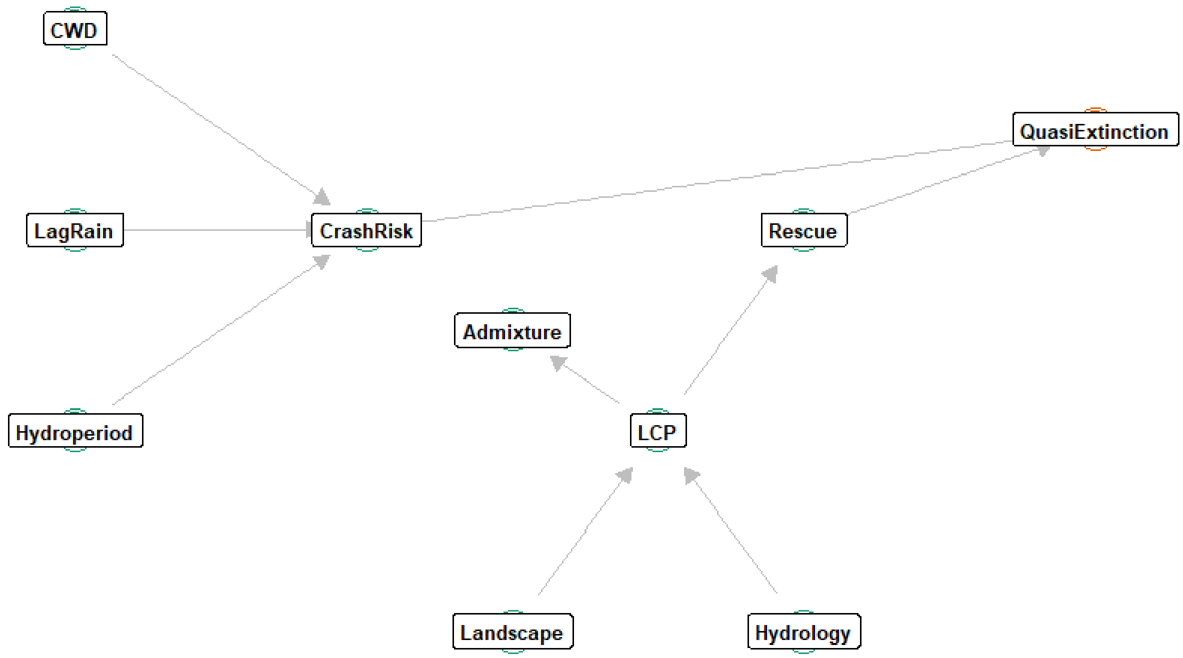
i. Decision Matrix

1. Different priorities based off 2024 EcoAdapt report

a. Ex. If we restore this specific riparian corridor, how much does regional quasi-extinction drop?

CRLF Predictive Metapopulation (DAG)

Streamlined framework isolating hydro-climate vulnerability and functional connectivity pathways.

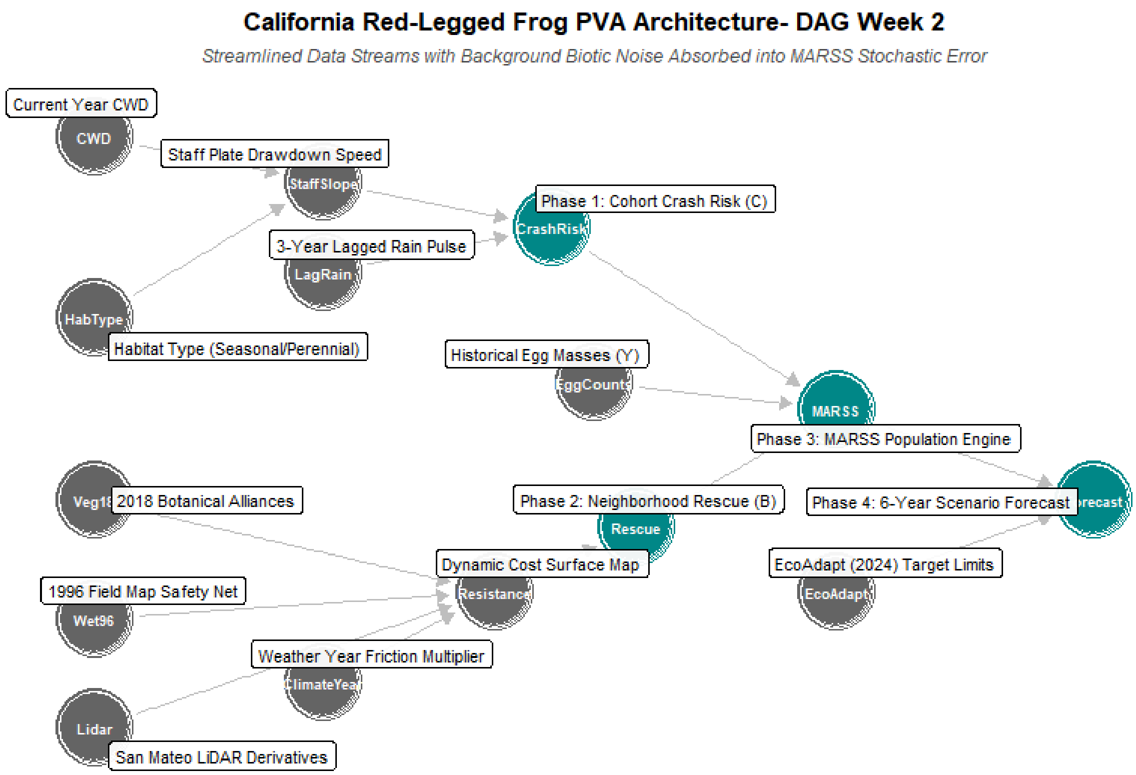



Notes

1. 10-M Kinda big
    a. Getting 2018 Lidar scans of Marin and San Mateo county
2. Different CWD in an area due to fogginess vs fogginess
3. Look at slope of when it gets to zero

# Week 3

June 15th, 2026




1.

2. Issues with CWD
    a. Currently missing data waiting on USGS
3. Cost surface map
    a. Focus on a given site - then compute cost to closest site or average of all neighbor sites
    b. Connectivity (opposite of cost) for a given site and year, since wet or dry years will differ in how connected they are -> that connectivity covariate that enters MARSS

# Week 4

Jun 22, 2026

1. Go over slides for data QA
    a. Egg Mass
        i. Distribution
    b. Staff water plates
        i. Anchor sites
        ii. Satellites Sites
        iii. Orphans Sites
2. Move around meetings for July

# Week 5

Jun 29, 2026

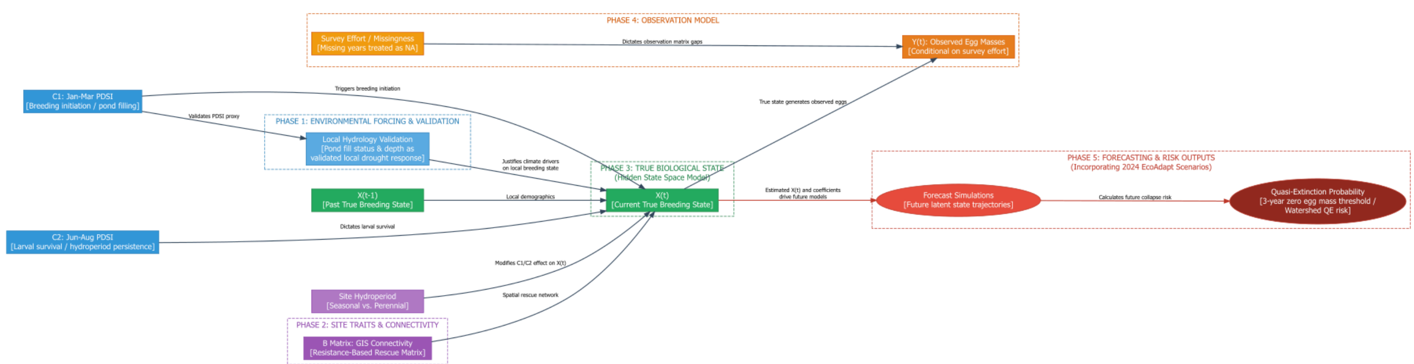
1. Rose & Halstead (2023) Identifying drivers of population dynamics for a stream breeding amphibian using time series of egg mass counts. (USGS / Ecosphere).
    a. What Their Model Looked Like: A Multivariate State-Space Multiple Population Viability Analysis (MPVA) built to separate human observation error from environmental process noise.
    b. Data Inputs: 30 years of messy, patchily sampled egg mass counts across 36 focal reaches, mapped against continuous climate drivers (precipitation extremes and streamflow).
    c. Their B and C Matrices: The C matrix tracked time varying hydrology and temperature drivers. Their spatial structure was restricted to a single stream channel network.
    d. Outputs Given: Quantified population growth rates, clear identification of hydroclimatic tipping points, and a clean isolation of observation error.
    e. Our Project: It serves as the primary defense for using egg masses as a legitimate proxy for true population states. We are adopting their exact state space logic but expanding the spatial network to a 2D pond landscape.
2. Stochastic metapopulation dynamics of a threatened amphibian to improve water delivery.
    a. What Their Model Looked Like: A spatially explicit, hydroecological stochastic metapopulation network model designed to simulate population vulnerability under climate escalated droughts.

b. Data Inputs: Wetland occupancy data paired with fine scale hydrologic inundation models.

c. Their B and C Matrices: The C matrix established sharp "interannual drying thresholds" for each wetland patch. The B matrix was structurally informed by spatial proximity to permanent water lifeboats to calculate recolonization probabilities.

d. Outputs Given: Future Quasi-Extinction Risk Curves (Cumulative Distribution Functions) showing the probability of network collapse over a multi-decade horizon under varying water management scenarios.

e. Our Project: It provides the blueprint for our future forecasting stage. Like Moor, we are evaluating how the spatial arrangement of permanent water assets "Anchors" protects vulnerable seasonal sites "Satellites" from crossing their climate driven drying thresholds (C2, June–August).




1. PHASE 1: Environmental Forcing & Validation

a. **Goal:** To establish a continuous, validated environmental driver that justifies using a regional climate index (PDSI) to predict local pond filling and breeding initiation.

b. **Data Inputs:** C1: Jan-Mar PDSI and raw localized hydrology records (staff plate depths and pond fill status).

c. **Completion Status: 100% Complete.**

d. **What was done:** Built the zero inflated GLMM to mathematically prove the relationship between PDSI and local depth, and used it to output the MARSS_Covariate_Matrix_C.csv. (72(sites) x 29(years)).

e. The fixed effect of regional winter climate (Jan–Mar Palmer Drought Severity Index; C1) was evaluated across two distinct sub models to account for the extreme zero-inflation present in the field data:

    i. The Conditional Depth Model (Pond Depth When Wet): Regional winter PDSI is a highly significant, positive predictor of physical water depth when a pond successfully fills (Beta= 0.02466, SE = 0.0092, Z = 2.682, p = 0.00732).

    ii. The Zero-Inflation Model (Probability of Remaining Bone Dry): Regional winter PDSI does not significantly alter the baseline probability of a pond remaining a structural zero (Beta = 0.06610, SE = 0.0612, Z = 1.080, p = 0.280).

    iii. While regional winter climate predictably controls how deep a pond fills (increasing depth by 2.5% per unit increase in PDSI), highly localized basin traits not regional macro-climates dictate whether a pond catches water or remains entirely dry.

2. PHASE 2: Site Traits & Connectivity

a. **Goal:** To define the static physical traits of the ponds and map the spatial network that allows frog populations to rescue each other via migration.

b. **Data Inputs:** Site Hydroperiod [Seasonal vs. Perennial] classifications and a GIS Resistance Raster to calculate the B Matrix: GIS Connectivity.

c. **Completion Status: ~50% Complete.**

d. **What was done:** Already successfully integrated the Site Hydroperiod traits when we created strict matching rules for our Anchor/Satellite/Orphan network.

e. **What is next:** Need to calculate the Resistance Based Rescue Matrix (the B matrix) using R spatial packages and LCP raster.

3. PHASE 3: Hidden State Space Model

a. **Goal:** To estimate the actual, unobserved frog population size—X(t) [Current True Breeding State] by mathematically combining their past population size X(t-1), localized demographic rates, and the environmental drivers.

b. **Data Inputs:** Past true states X(t-1), the validated Phase 1 environmental forcing matrix, the Phase 2 B matrix, and C2: Jun-Aug PDSI (which dictates larval survival and hydroperiod persistence).

c. **Completion Status:** 0% Complete. (But 100% prepared).

d. **What was done:** Built the required inputs, but we haven't executed the math yet.

e. **What is next:** Once Phase 2 is finished, I will feed the matrices into the MARSS in R to actually run the state-space algorithm and extract the underlying population estimates.

4. PHASE 4: Observation Model

a. **Goal:** To link the true hidden biological state X(t) to what was actually observed in the field Y(t), explicitly accounting for imperfect detection and missing surveys.

b. **Data Inputs:** The true biological state X(t) generated by Phase 3, and Survey Effort / Missingness rules.

c. **Completion Status:** 100% Complete.

d. **What was done:** By building MARSS_Observation_Matrix_Y.csv and intentionally leaving missing survey years as NA, we explicitly encoded the "Survey Effort" into the model structure.

e. **What is next:** Nothing to prep. The MARSS algorithm will automatically handle the NA values during the Phase 3 run, conditioning the observation matrix Y(t) on those missingness rules.

5. PHASE 5: Forecasting & Risk outputs

   a. **Goal:** To use the newly calculated parameters from Phase 3 to run simulations of future population trajectories and calculate the risk of localized extinction.

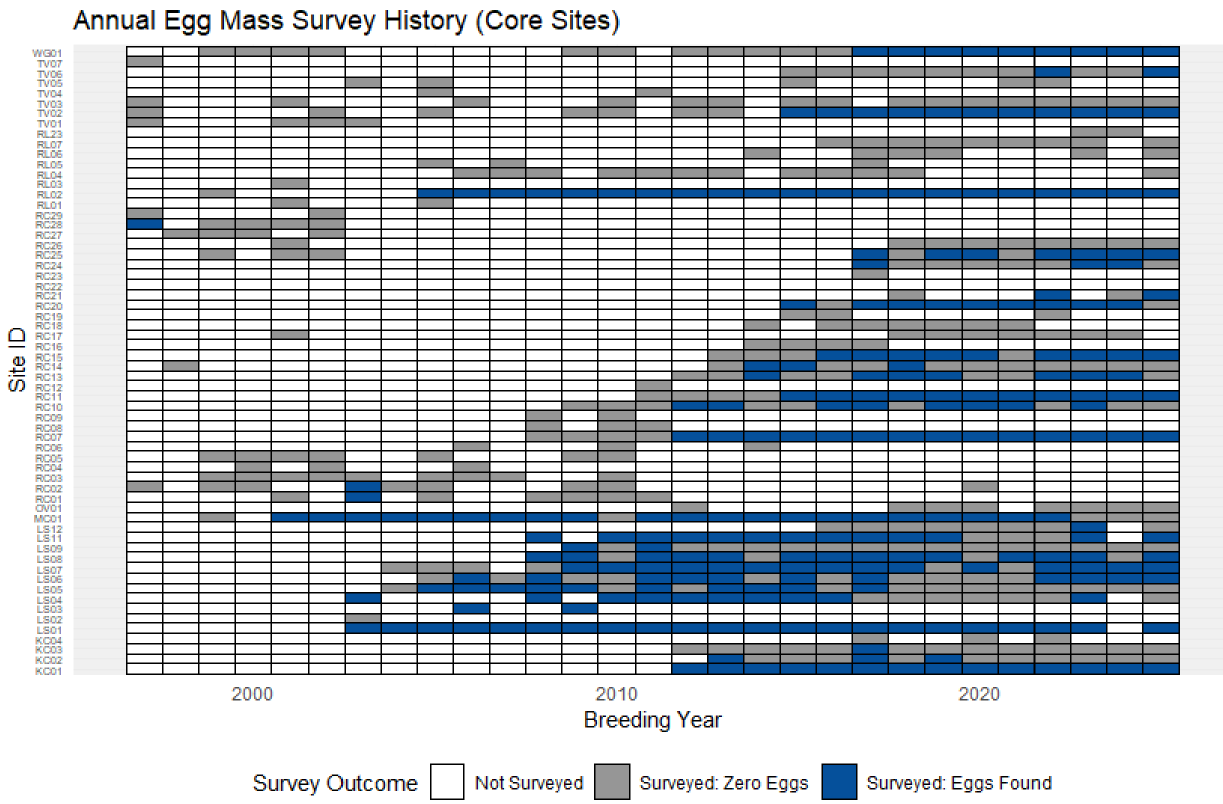
   b. **Data Inputs:** The estimated parameters and coefficients from Phase 3, combined with future climate projections via 2024 EcoAdapt Scenarios.

   c. **Completion Status:** 0% Complete.

   d. **What is next:** After Phase 3 gives us the final model coefficients, we will project X(t) into the future to calculate Quasi-Extinction Probability, utilizing specific thresholds (like 3 consecutive years of zero egg masses).

# Week 6

July 8, 2026




1.

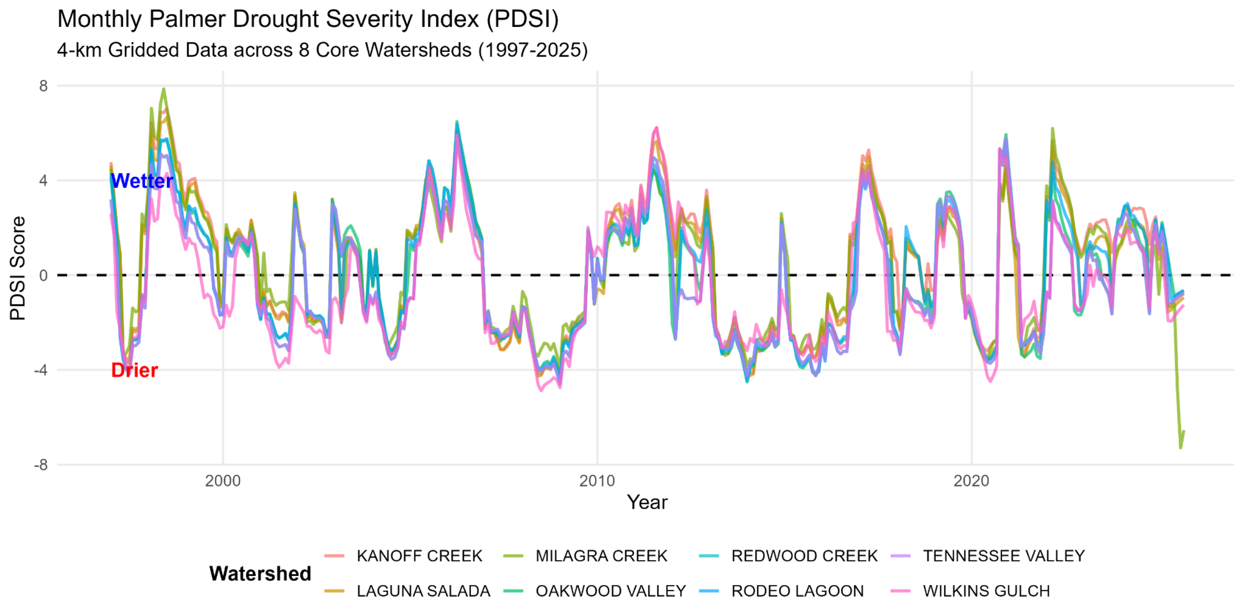
2. Site that change ID number

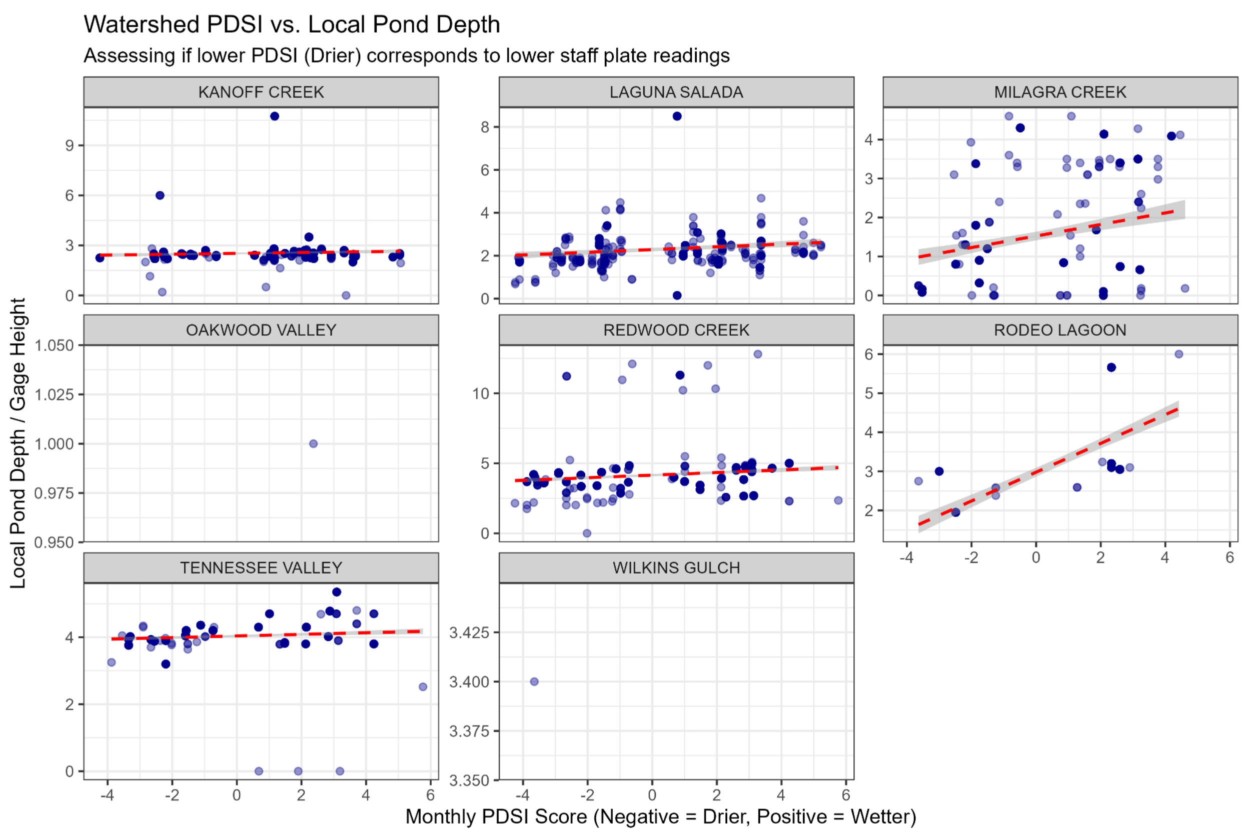
3. Look at Drough index vs continuous data and compare them spearman correlations

    a. Minimum more important for biology

    b. Minimum depth for that month vs its pdsi correspondent

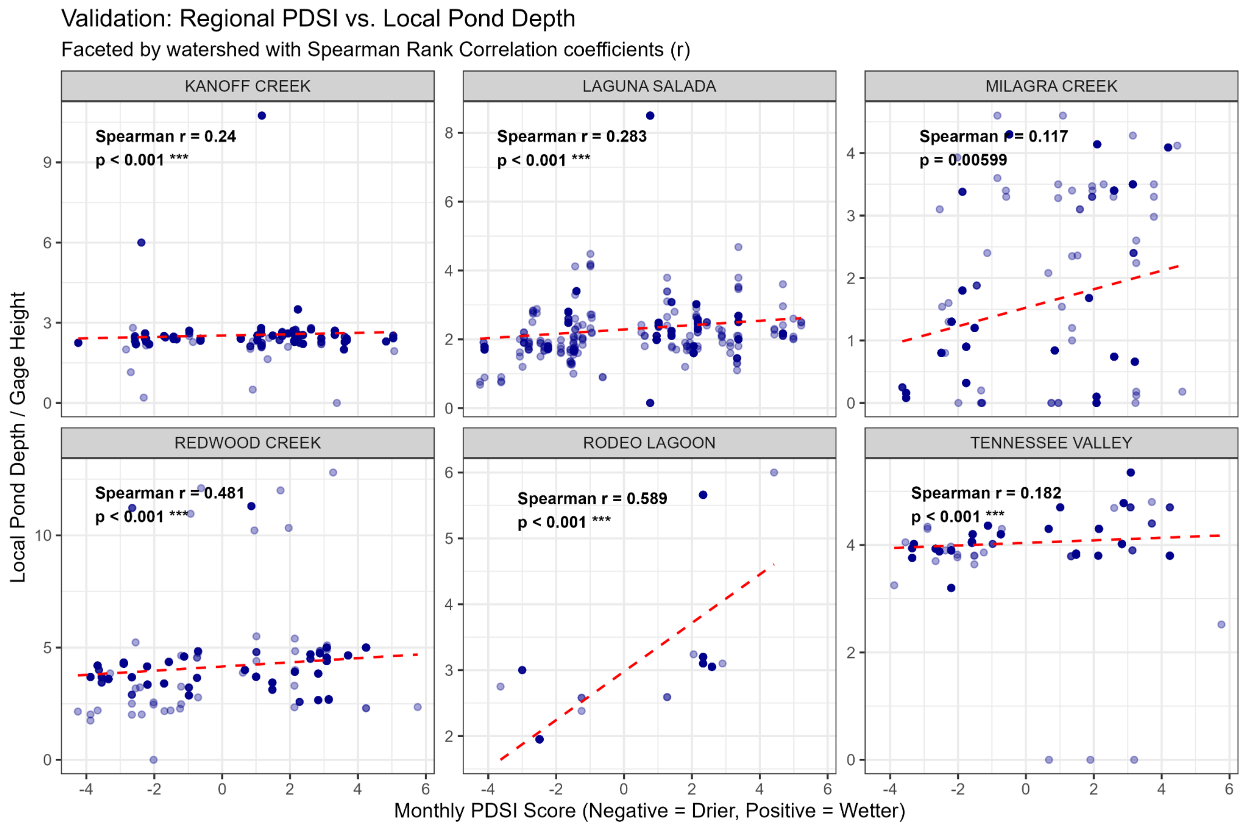
4. Completeness of the Sites

1. 
Monthly Palmer Drought Severity Index (PDSI)
4-km Gridded Data across 8 Core Watersheds (1997-2025)
PDSI Score
Wetter
Drier
8
4
0
-4
-8
2000
2010
2020
Year
Watershed
KANOFF CREEK
LAGUNA SALADA
MILAGRA CREEK
OAKWOOD VALLEY
REDWOOD CREEK
RODEO LAGOON
TENNESSEE VALLEY
WILKINS GULCH

2. 
Watershed PDSI vs. Local Pond Depth
Assessing if lower PDSI (Drier) corresponds to lower staff plate readings
KANOFF CREEK
LAGUNA SALADA
MILAGRA CREEK
OAKWOOD VALLEY
REDWOOD CREEK
RODEO LAGOON
TENNESSEE VALLEY
WILKINS GULCH
Local Pond Depth / Gage Height
Monthly PDSI Score (Negative = Drier, Positive = Wetter)

3.

| watershed | n_readings | spearman_rho | p_value | significance |
| --- | --- | --- | --- | --- |
| KANOFF CREEK | 985 | 0.24 | 2.42E-14 | *** (Highly Significant) |
| LAGUNA SALADA | 694 | 0.283 | 2.85E-14 | *** (Highly Significant) |
| MILAGRA CREEK | 554 | 0.117 | 0.00599 | ** (Very Significant) |
| REDWOOD CREEK | 822 | 0.481 | 6.85E-49 | *** (Highly Significant) |
| RODEO LAGOON | 306 | 0.589 | 6.54E-30 | *** (Highly Significant) |
| TENNESSEE VALLEY | 685 | 0.182 | 1.60E-06 | *** (Highly Significant) |

4. 

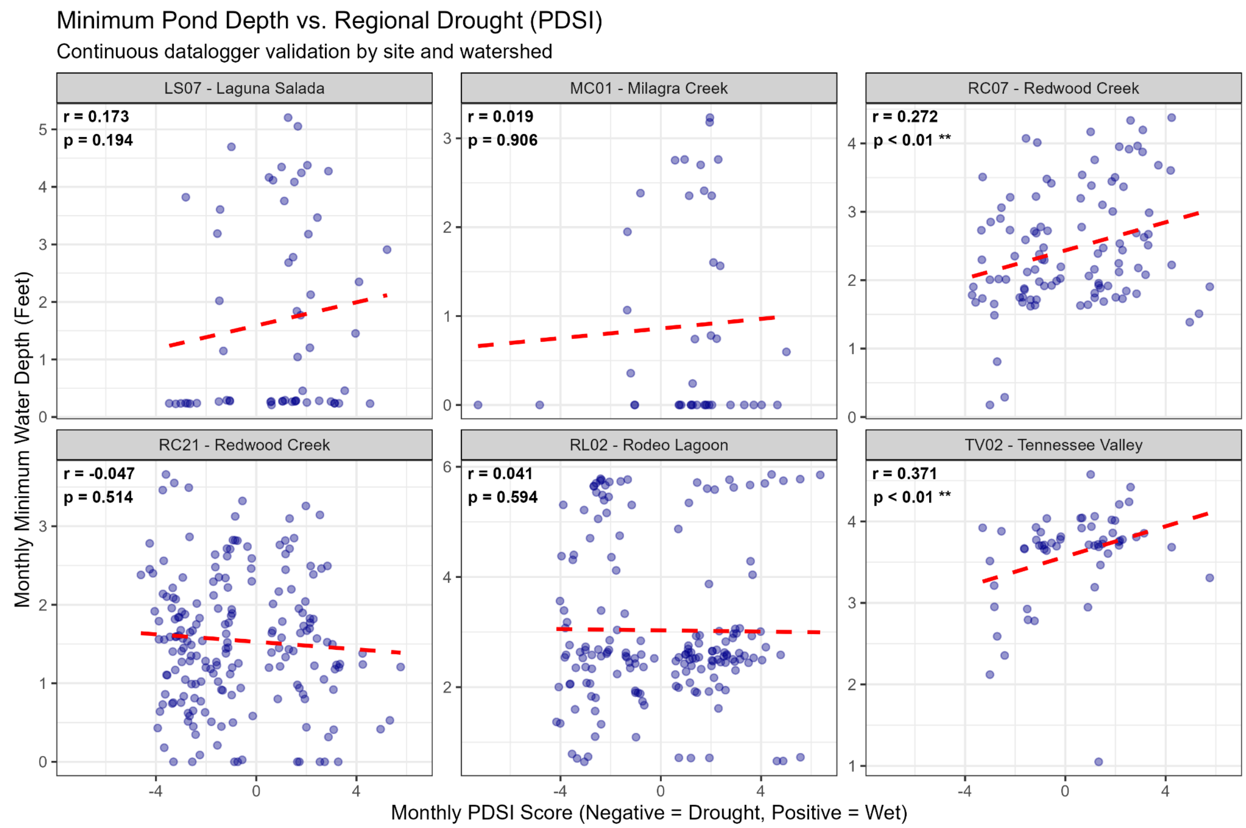

5. Spearman values are low
   a. Max pond depths? Cant get any lower than 0ft and cant get any higher than its own spillway?
6. Next Steps
   a. Work on Zero-Inflated Model
      i. MARSS needs the environmental covariate matrix at 100% complete
      ii. To fill in missing years and unmonitored sites
         1. Use regional PDSI data to interpolate the missing local pond depths
         2. Need to account for inflated number of 0's
7. Darren Updates
   a. Frogs mostly breed in the winter months

        i. Take 4-6 months to mature
            1. 4 months: 50% Frog conversion rate
            2. 6 months: 90% Frog conversion rate
        ii. The longer a pond holds water the more successful it
    b. Need to check Muir Beach ponds in the Redwood Creek Watershed
        i. Replaced ID's need to find old and new and merge the time series
        ii. Wetlands are no longer wetlands due to habitat changes
            1. Consistent surveyed areas that were areas for breeding frogs
            2. Polygons change over time
            3. Get locations for the old and new ponds
    c. Continuous Water Depth Data for certain sites
        i. Around half or full hour reports of data
        ii. Some data gaps due to battery life of machine
        iii. Connect dots of missing water data during the dry season
        iv. Before I forget: Do we want specific sites or just for sites with longest records?

8. Meeting Notes
    a. PDSI time series try to break up watershed by county
    b. Slope per pond site in Watershed
    c. Milagra Creek PDSI is a bit odd need to look at that more

# Week 7

Jul 17, 2026




1.

2.

   a. Where do we want to go with this here?
   b. Model ideas
      i. PDSI not correlated at this scale
      ii. Models:
         1. PDSI is proxy for temp and theoretically pond status. T-3 = 3 year lag for female sexual maturity
         2. DependentVariable = if there is reproduction…not necessarily recruitment.
         3. EggCount ~ # use counts as in traditional MARSS
            mean(PDSI$_{\text{Oct-Dec or Jan}}$) * # *early season conditions*
            *Save both Oct-Dec and Oct-Jan to make it easy to switch from Oct-Dec or Oct-Jan for later*
            mean(PDSI$_{\text{Feb-Mar}}$) + # late season conditions
            depth + # data very limited, maybe can't use
            PondType + # Seasonal/perennial
            Connectivity + # working on LCP analysis
            mean(PDSI$_{\text{Oct-Dec or Jan(t-3)}}$) * # lag effect data for breeding success from 3 yr prior.
            mean(PDSI$_{\text{Feb-Mar(t-3)}}$) # lag effect

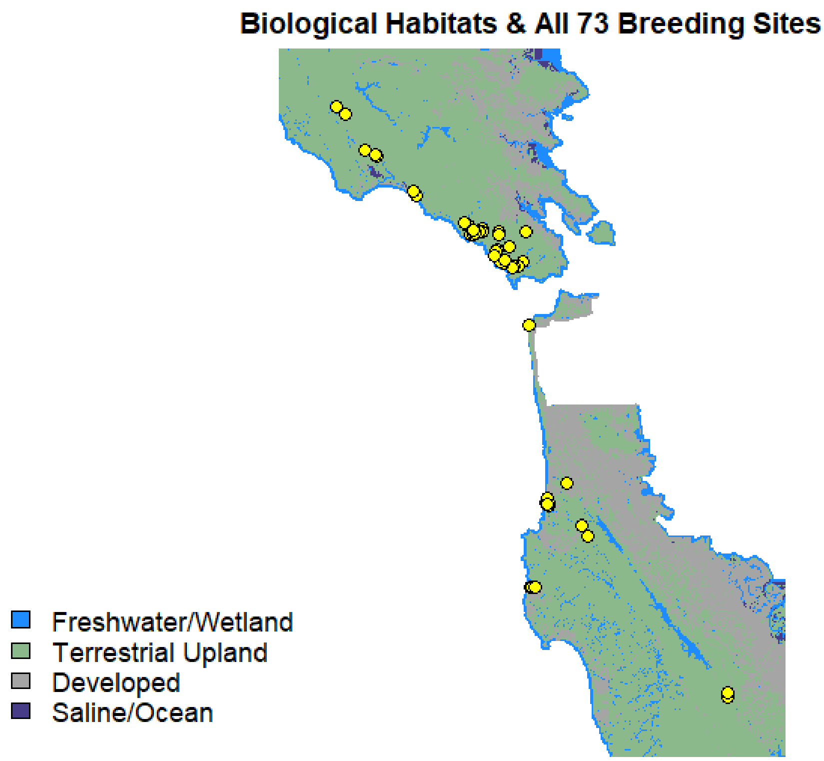
3a. Perhaps try to build the MARSS model without depth for now.

Extinction probabilities by site for existing data.

3b: HYPOTHESES:

Write out the hypotheses for each variable or interaction before running the model.

4. Lag-Variable Options
    a. mean($PDSI_{\text{Oct-Dec or Jan(t-3)}}$) * mean($PDSI_{\text{Feb-Mar(t-3)}}$) + ←optionA
    $Eggcount_{s,\ t\text{-}3}$ ← Option B
5. Stranded egg mass categories
    a. New egg mass status
    b. Then subsequent surveys - note if stranded.
    c. How use this data?
6. Regression for staff plate depth data
    a. Staff plate data derived from nearby ponds
        i. but WG02 has no pairs
        **ii. So Lump WG with RW creek**
            1. RC07 has both staff plate and continuous are mostly perennial sites.
            2. Lagoon = perennial never goes dry
            3. In RC All other sites- seasonal.




3.

4.

| Habitat_Category | Map_Color | Landscape_Features |
|---|---|---|
| Freshwater / Wetland | Deep Blue | Ponds, creeks, marshes, and Marin Reedgrass. |
| Terrestrial Upland | Muted Green | Grasslands, scrub, and dry land. |
| Developed / Roads | Gray | Highways, paved roads, and urban buildings. |
| Saline / Ocean | Dark Blue | Open ocean, SF Bay, and salt flats. |

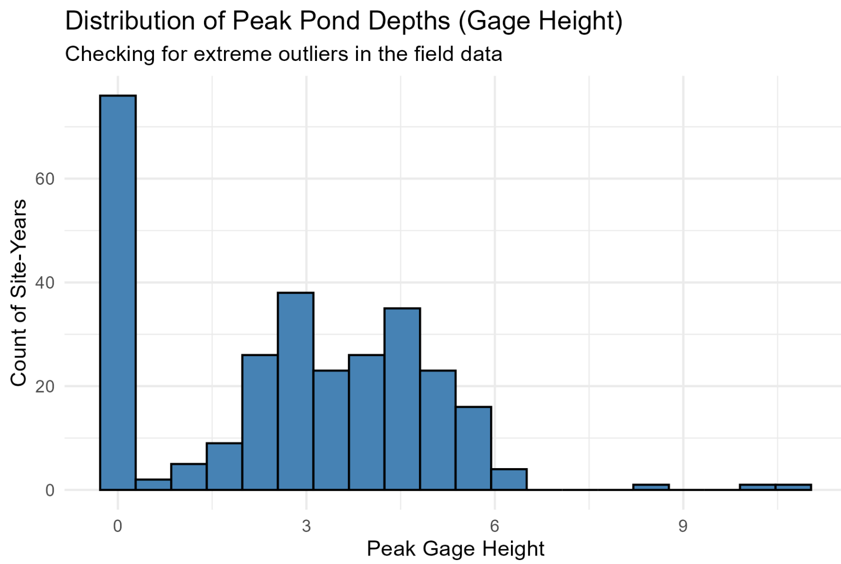
5. 2018 Fine Scale Vegetation Map
    a. 10M x 10M Resolution
        i. Vegetation/Land Cover
        ii. DEM & Slope
        iii. Breeding Sites
            1. PLAN TO ADD
            2. Cost to each vegetation classification
            3. PDSI multiplier
                a. More Budget in Wet
                b. Less Budget in Dry
    b. Divided into 4 groups that are going to be our main fields to transverse
        i. Freshwater/Wetland
            1. Prime Habitat
                a. Least Cost
        ii. Terrestrial Upland
            1. Habitable
                a. Less Cost
        iii. Developed
            1. Roads/Pavement/Development
                a. More Cost
        iv. Saline/Ocean
            1. Saline
                a. Highest Cost

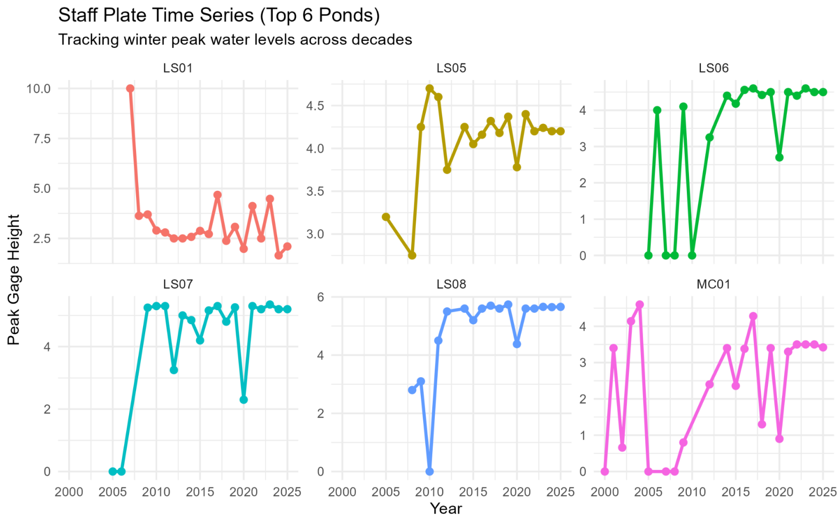
i. Will kill frogs

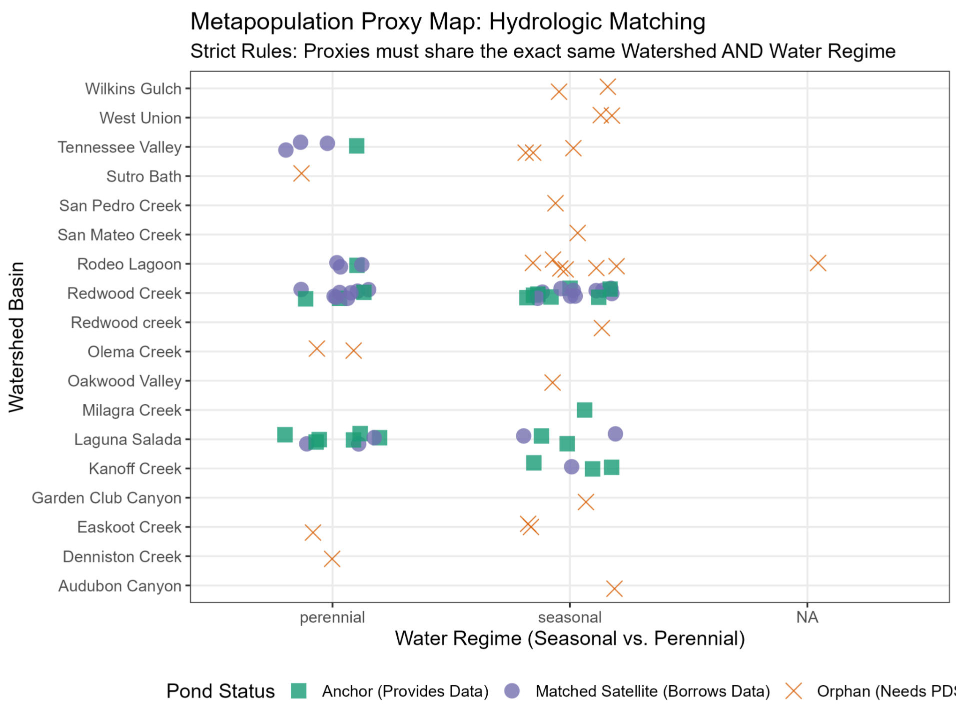
6. Dynamic Cost Surface Model
    a. LCP friction does not change
    b. DCSM friction changes due to adding climate to the equation
        i. Dry Years more cost to do move
        ii. Wet Years less cost to do move
    c. Need to assign cost to each classification

Meeting Notes

1. Any memory or lag in PDSI?
2. Maybe look at yearly instead of monthly
    a. Accumulation of the season instead of each month
3. Continuous data and staff plate data
4. Time between median of breeding and when pond dries compare to what we think we know is needed for successful metamorphosis
    a. Frogs mostly breed in the winter months
        i. Take 4-6 months to mature
            1. 4 months: 50% Frog conversion rate
            2. 6 months: 90% Frog conversion rate
        ii. The longer a pond holds water the more successful it
5. Regression for each year to compare against the continuous data with staff plate
6. Building larger regression model with the ponds classified by water type different slopes in relationship to PDSI
7. Probability of ponds persisting throughout the year whether its seasonal or year long


Distribution of Peak Pond Depths (Gage Height)
Checking for extreme outliers in the field data
Count of Site-Years
0
20
40
60
0
3
6
9
Peak Gage Height




Scatterplot for each watershed
Look for Oakwood valley staff plate data again

- Borrow depths across anywhere in Marin
- But in San Mateo county borrow closest possible watershed?
    - L

- Compare but the staff plate value is changing over time oct-march.

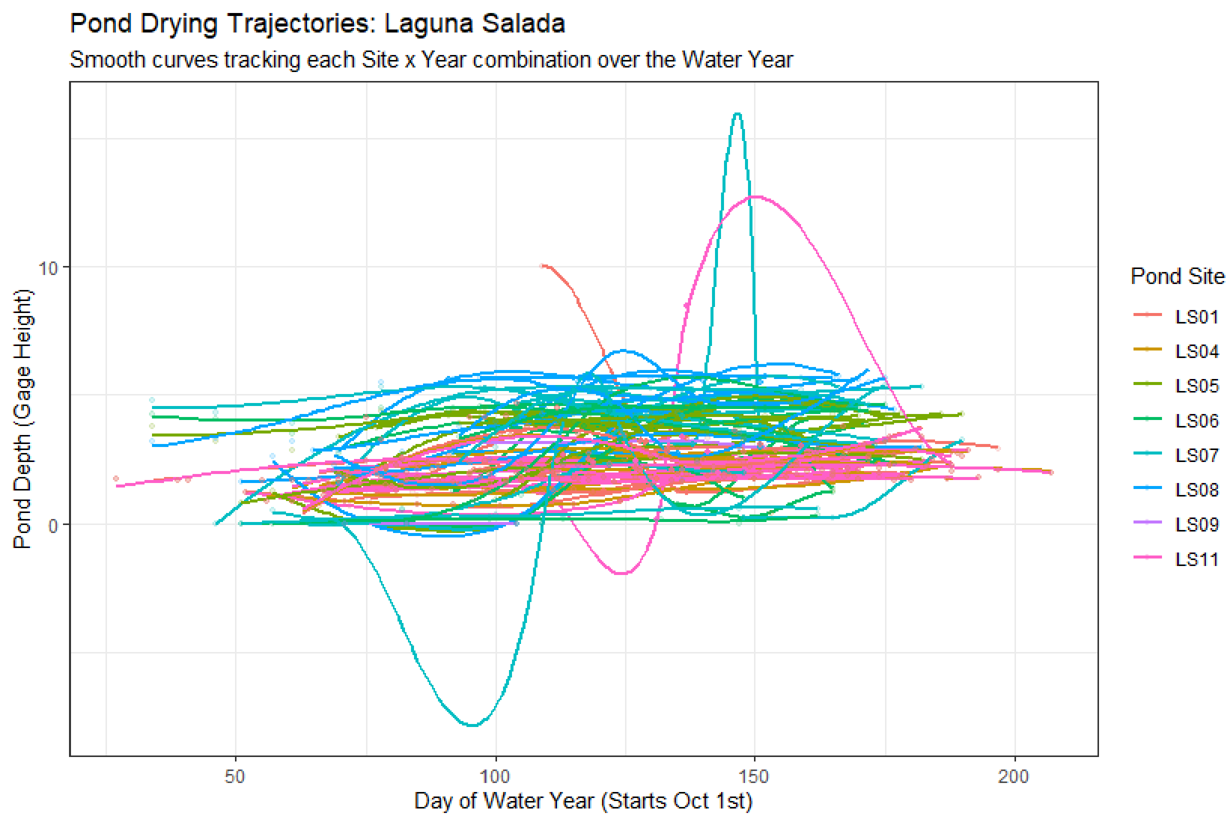
Multiple scatterplot for correlation analysis: How well Sites correlate to each other in the same watersheds. X axis is date of staff plate entry and Y axis is depth on the day it was recorded each line  on the plot is going to be year each year will proceed differently do different sites follow use all the staff plate data we got

Multiple scatterplot for correlation analysis: How well watersheds correlate between sites with long term staff plate data with anchor sites. Watershed vs Watershed

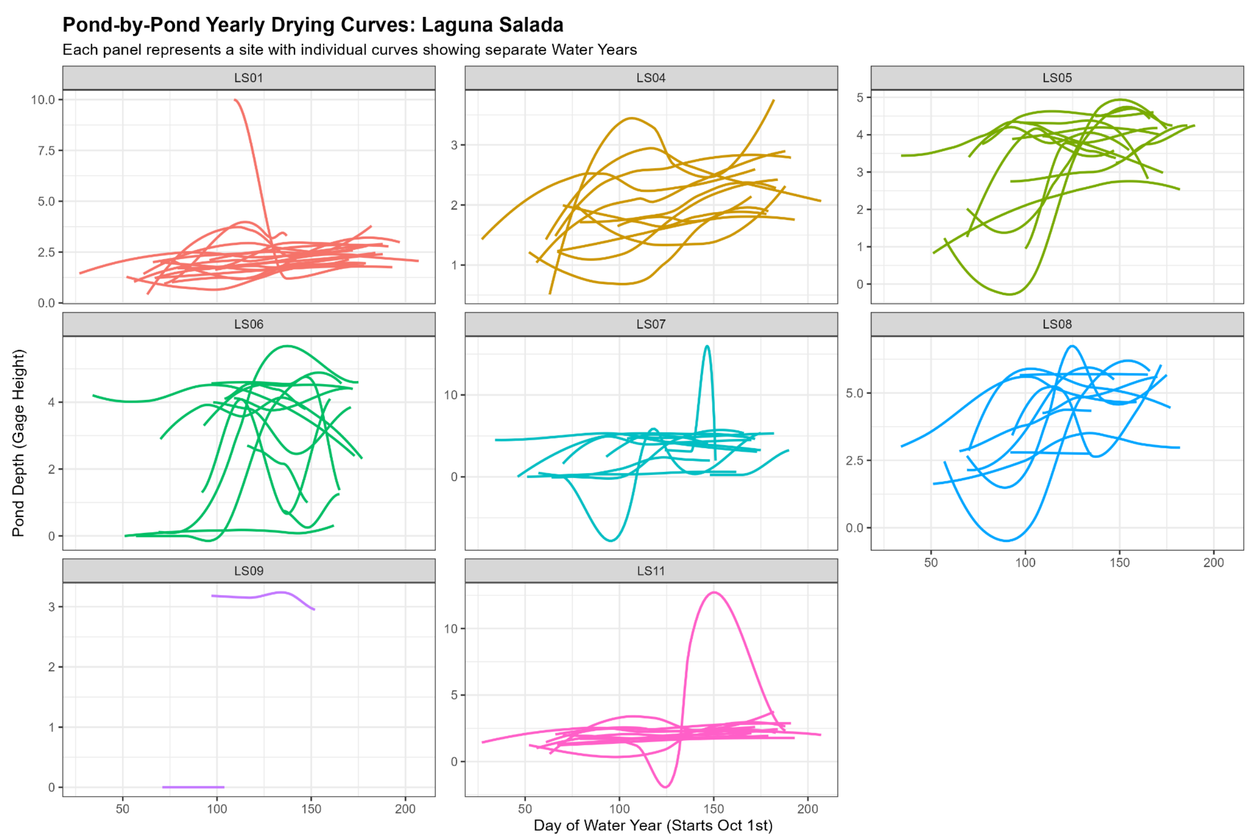
Jul 16, 2026

1. Ben & Anthony Check in
    a. x = day of water year (OCT 1-SEP30)
    b. y = depth
    c. each line is a different site x year combination.  we should expect a negative slope for each line, but want to know if the slopes are similar for each line.
    d. use a different color for each site
    e. smooth lines that follow each site x year combination
    f. this would produce many lines on a single plot.  9 sites by x years.

2.


Pond Drying Trajectories: Laguna Salada
Smooth curves tracking each Site x Year combination over the Water Year
Pond Depth (Gage Height)
10
0
50
100
150
200
Day of Water Year (Starts Oct 1st)
Pond Site
LS01
LS04
LS05
LS06
LS07
LS08
LS09
LS11

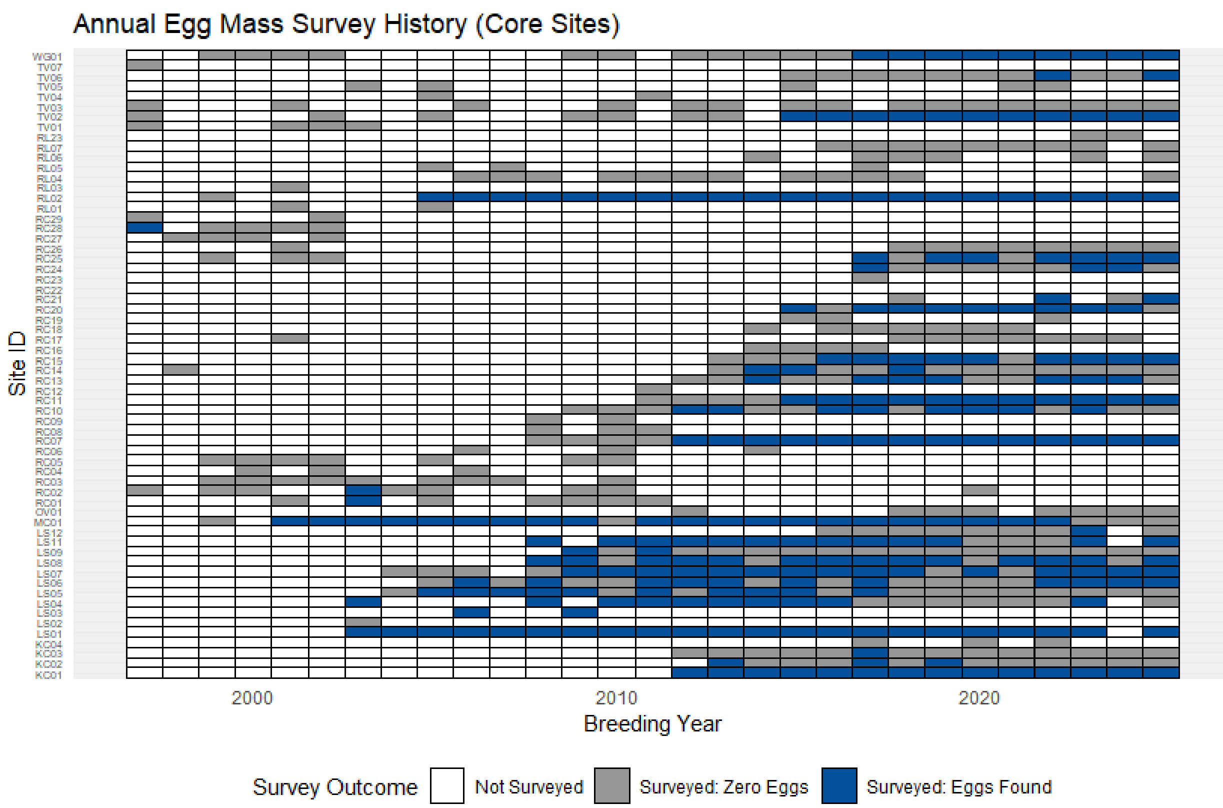
3.


Pond-by-Pond Yearly Drying Curves: Laguna Salada
Each panel represents a site with individual curves showing separate Water Years
LS01
LS04
LS05
LS06
LS07
LS08
LS09
LS11
Pond Depth (Gage Height)
Day of Water Year (Starts Oct 1st)

4. Geom smooth and geom data point span argument increase span to make them less wiggly method =linear model
5. Model to test for similar slopes in depth change across sites. Run for each watershed separately.
    a. `WatershedDepthModel <- lm(depth ~ site * DayOfWaterYear * StreamType + (1|Year), data = DATA)`
    b. `Can also Try run all watersheds:`

       `AllWatershedDepthModel <- lm(depth ~ site * DayOfWaterYear * StreamType + Watershed + (1|Year) + (1|WaterShed), data = DATA)`

    c.




6.
7. Talking points with Albert

a. Actual output for the extinction model
b. Some modeling set up without depth while we figure that out
c. Talk about the framework
d. Map out the framework for the modeling

Week 7.5

Jul 17, 2026

1. Need to summarize the Extinction model papers
    a. Rose et al (2023). Egg count as population proxy
    b. Smits et al (2019). PDSI as covariate in state equation
    c. MARSS R User Guide. Chapter 18 Biological Lag
        i. Expanded state vector to handle T-3 maturity
        ii. Lag-p instead of lag-1
        iii. Xt =[xt,xt-1,xt-2]
            1. X^1t-3
    d. Holmes et al (2007) CSEG simulations to find quasi-extinction probability
        i. Corrupted Stochastic Exponential Gaussian
    e. Some of the code they use
    f. How we plan to use it for our framework.
2. EggCount ~ # use counts as in traditional MARSS

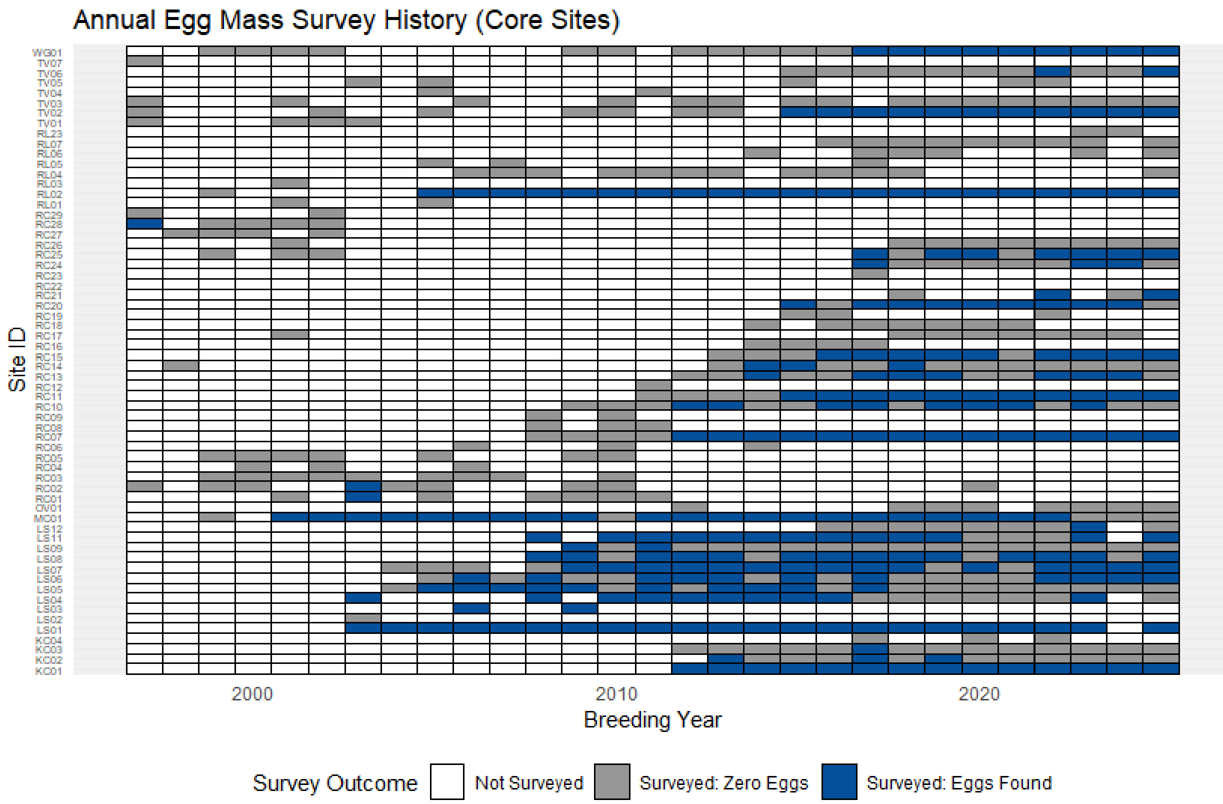
    I. mean($PDSI_{Oct\text{-}Dec}$) * # early season conditions
    II. mean($PDSI_{Jan\text{-}Mar}$) + # late season conditions
    Oct-jan feb-march
    III. depth + #
    IV. PondType + # Seasonal/perennial (Site clarification not covariate)
    V. Connectivity + # working on LCP analysis
    VI. mean($PDSI_{Oct\text{-}Dec(t-3)}$) * # lag effect data
    VII. mean($PDSI_{Jan\text{-}Mar(t-3)}$) # lag effect

Variates (response variable): EggCount
- Dimensions of the matrix:
    - 72 sites [fewer after filtering] x 29 years (72 rows x 29 columns) <- PLOT IT

Covariate (explanatory variable:
Mean (PDSI) Oct-Dec current year + t-3 (Model 1) <- PLOT IT
- 6 PDSI (one per watershed) x 29 values (one per year)
Mean (PDSI) Jan-Mar current year + t-3 (Model 2) <- PLOT IT

- 6 PDSI (one per watershed) x 29 values (one per year)

3a. Perhaps try to build the MARSS model without depth for now.
Extinction probabilities by site for existing data.




1.
2. Sites that have 15 ANNUAL values or more 0 is a value (meaning that they've been monitored at least half of the time)
    a. Plot of the sites that meet the the threshold

Variates (response variable): EggCount
- Dimensions of the matrix:
    - 72 sites [fewer after filtering] x 29 years (72 rows x 29 columns) <- PLOT IT

Covariate (explanatory variable:
Mean (PDSI) Oct-Dec current year + t-3 (Model 1) <- PLOT IT
- 6 PDSI (one per watershed) x 29 values (one per year)

Mean (PDSI) Jan-Mar current year + t-3 (Model 2) <- PLOT IT
- 6 PDSI (one per watershed) x 29 values (one per year)

3a. Perhaps try to build the MARSS model without depth for now.
Extinction probabilities by site for existing data.

Each site with data becomes its own state Q should be diagonal and equal each pond should have different levels of observations
Potential model structures:
Z (connects observations to states) -> “identity”
R = “zero” (so it’d be MAR rather than MARSS)
B = “identity”
U = “unequal” (long-term trend)
Q = “diagonal and unequal” (allow each pond to have different levels of variability)

After we see if it works, then we add covariates

3. Hopeful output for extinction model?
    a. Percentage of future simulation paths in which we meet the critical defined thresholds (Ex. 3+ years of consecutive no egg masses per site) over a time frame (EX. 15-35 years etc.)
    b. 95% and 50% confidence intervals on risk estimates.
    c. Population Baseline Eggmass count growth
        i. Average growth rate for different pond types
            1. Perennial vs Seasonal
    d. C-Matrix (Coefficient effects on egg production)
        i. Early PDSI
            1. Oct-Dec or Jan
        ii. Late PDSI
            1. Jan-Mar or Feb-Mar
        iii. Connectivity
            1. For each pond/site in each year
                a. How connected it is to the rest of the pond network through the landscape?

b. Site-year connectivity/rescue score

iv. Observation Error Variance ( R )

1. Human Error in collection of data

v. Process Variance ( Q )

1. True Ecological Variance

4. State Equation

a. Xt = Bxt-1 + U + Cct + Wt

i. Xt = True population

1. Hidden state population
2. Keep in mind the t-3 lag

ii. Bxt-1 = Density Control

1. Is population stable or not after disturbance
2. Xt-1 must be multiplied by previous state to act as a transition state
3. Keep in mind the t-3 lag

iii. U = Growth Trend

1. Up or Down for pond type

iv. Cct = Climate Impacts

1. How much these covariates impact. The weight of it

v. Wt = Biological variance

1. Random ecological happenings

5. Observation Equation

a. Yt = Zxt + A + Vt

i. Yt = Raw Data

ii. Zxt = Mapping Matrix

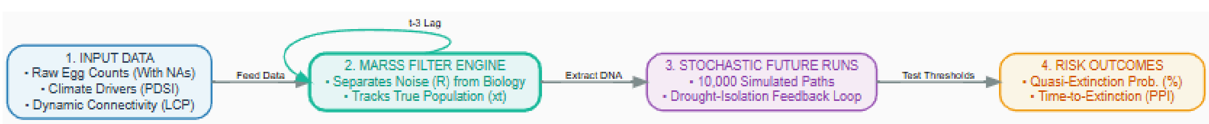
1. Links field count to hidden population state

iii. A = Bias

iv. Vt = Observation error

1. Human error when collecting data

6. OUTPUT Fitted Parameters
    a. C Matrix
        i. Direct impact score for each variable early PDSI late PDSI or Connectivity on frog population
    b. U vector
        i. Identify underlying growth or decline momentum of each pond independent of climate spikes
    c. Q and R
        i. Reliability scores of how much of population flux are due to Q or R
            1. Q = Environmental Schock
            2. R = Human Error
7. Quasi-extinction modeling
    a. Where we get the percentage of potential futures where the sites reach the extinction thresholds (EG. 3+ consecutive years of no egg masses)

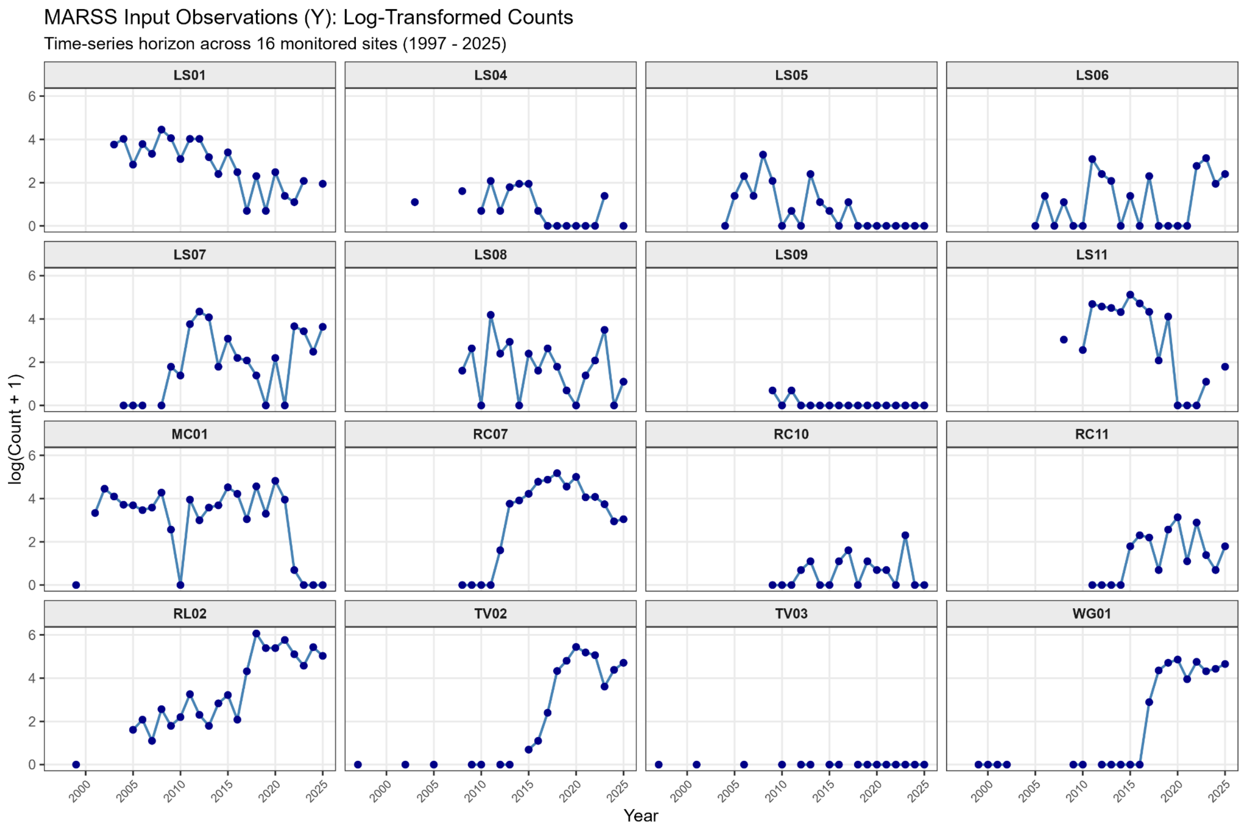



8.

# Week 9

Jul 27, 2026

1.




Had to have 15+ years of data
Originally had 16 sites but 2 of them were all or mostly no egg masses surveyed LS09 & TV03 (DF- yes just checked and confirming that those sites TV03-no eggs and LS09-1 or 2 eggs)

Potential model structures:
Z (connects observations to states) -> “identity”
R = “zero” (so it’d be MAR rather than MARSS)
B = “identity”
U = “unequal” (long-term trend)
Q = “diagonal and unequal” (allow each pond to have different levels of variability)
C = Sites will only go to their watershed LS laguna salda Ls WG 0

After we see if it works, then we add covariates

Covariates

Check for correlation for between current year PDSI and T-3 Pearson

Add a legend for the U and Q parameters in simple terms

Maybe try R as not 0

Model 1: Oct-Dec + Oct-Dec T-3

New Model 1: Oct -Dec (Early)

New Model 1.5: Oct-Dec T-3 (Early lag)

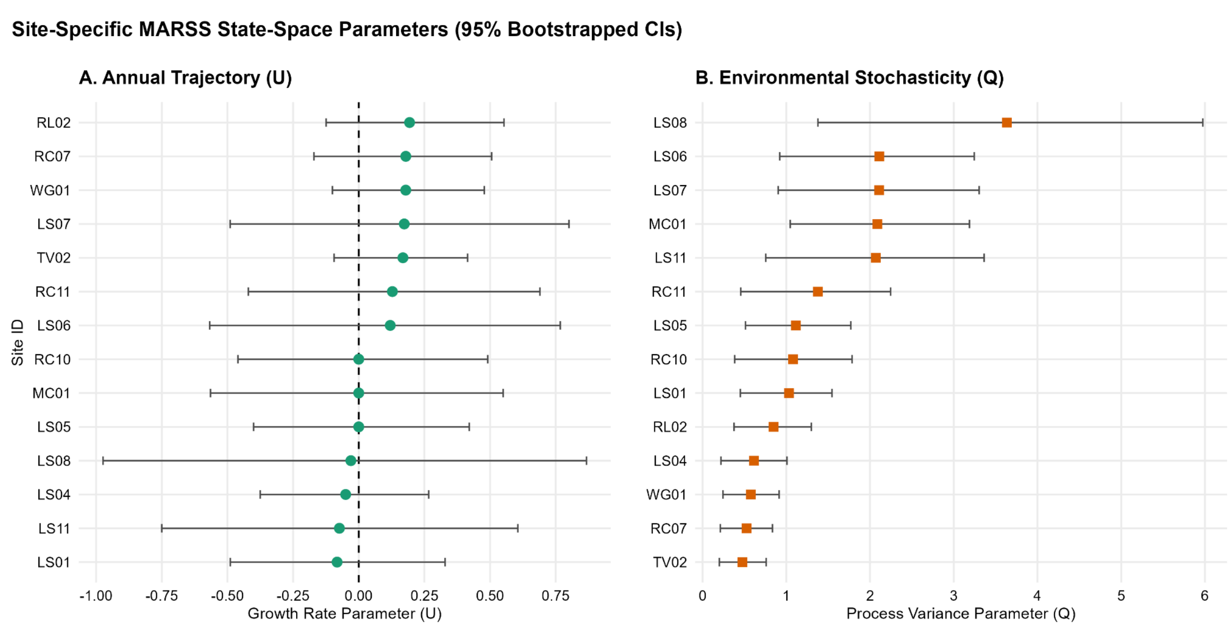
Model 2: Jan-March + Jan-March T-3

New Model 2: Jan-March (Late)

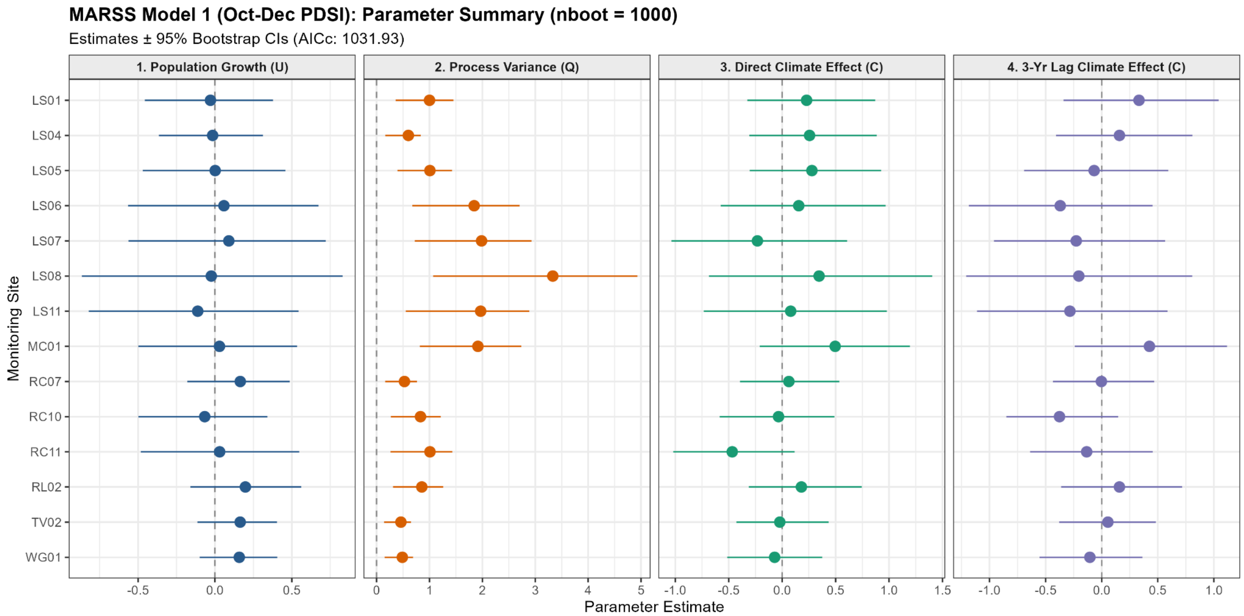
New Model 2.5: Jan-March T-3 (Late lag)

Z (connects observations to states) -> "identity"
R = "zero" (so it'd be MAR rather than MARSS)
B = "identity"
U = "unequal" (long-term trend)
Q = "diagonal and unequal" (allow each pond to have different levels of variability)
C = "Custom" Pond only responds to PDSI in its own watershed otherwise 0

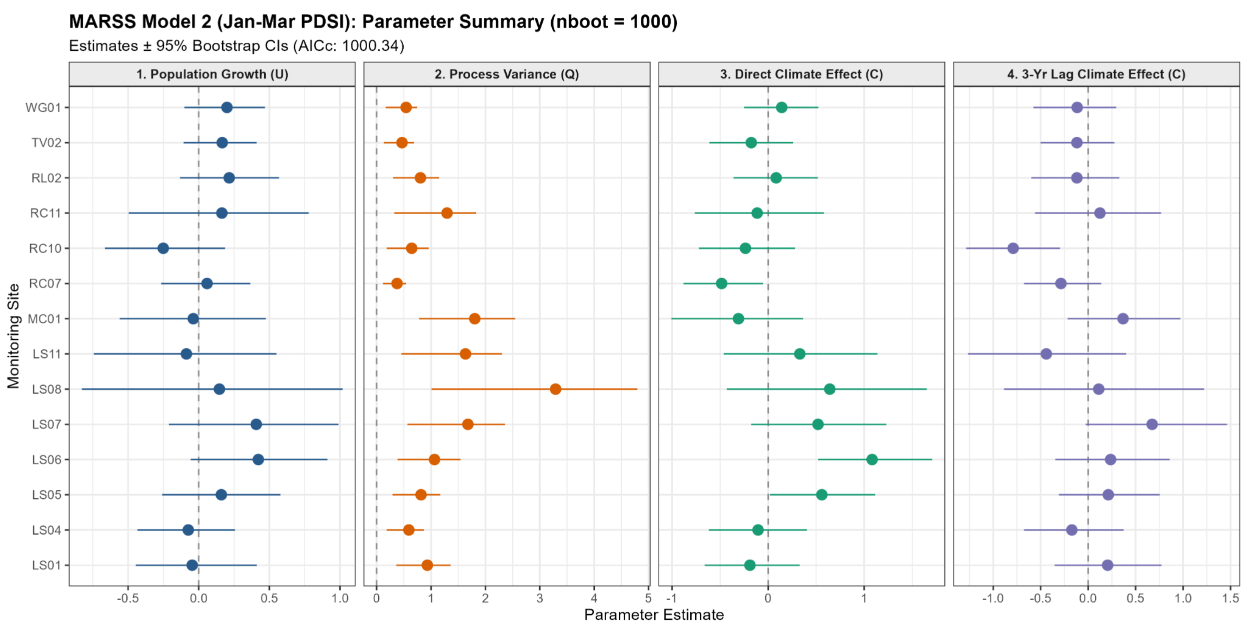
Baseline Model no covariates




Model 1: Oct-Jan + Oct-Jan T-3




Model 2: Jan-March + Jan-March T-3




Model 2: 4 non zero results

LS05: +0.559 wetter conditions in Jan-March associated with increase in egg production

LS06: +1.082 strong correlation associated with egg production in wetter Jan-March

RC07: -0.484 wetter conditions in Jan-March associated with decrease in egg production

RC10: -0.790 wetter conditions 3 years earlier in Jan-March associated with decrease in eggs

Comparison

| U | Mean U | Range U | # of sites with 95%CI excluding Zero |
|---|---|---|---|
| Baseline: | 0.065 | -0.083 to 0.194 | 0/14 |
| Model 1(Oct-Dec + t-3): | 0.046 | -0.112 to 0.198 | 0/14 |
| Model 2(Jan-March + t-3): | 0.103 | -0.251 to 0.422 | 0/14 |

| Q | Mean Q | Median Q | Range Q |
|---|---|---|---|
| Baseline: | 1.404 | 1.097 | 0.475 to 3.635 |
| Model 1(Oct-Dec + t-3): | 1.273 | 1.004 | 0.460 to 3.331 |
| Model 2(Jan-March + t-3): | 1.139 | 0.875 | 0.376 to 3.290 |

| Parameter | Baseline | M1: Oct-Dec | M2: Jan-March |
|---|---|---|---|
| Supported Current C | NA | 0/14 | 3/14 |
| Supported C t-3 | NA | 0/14 | 1/14 |

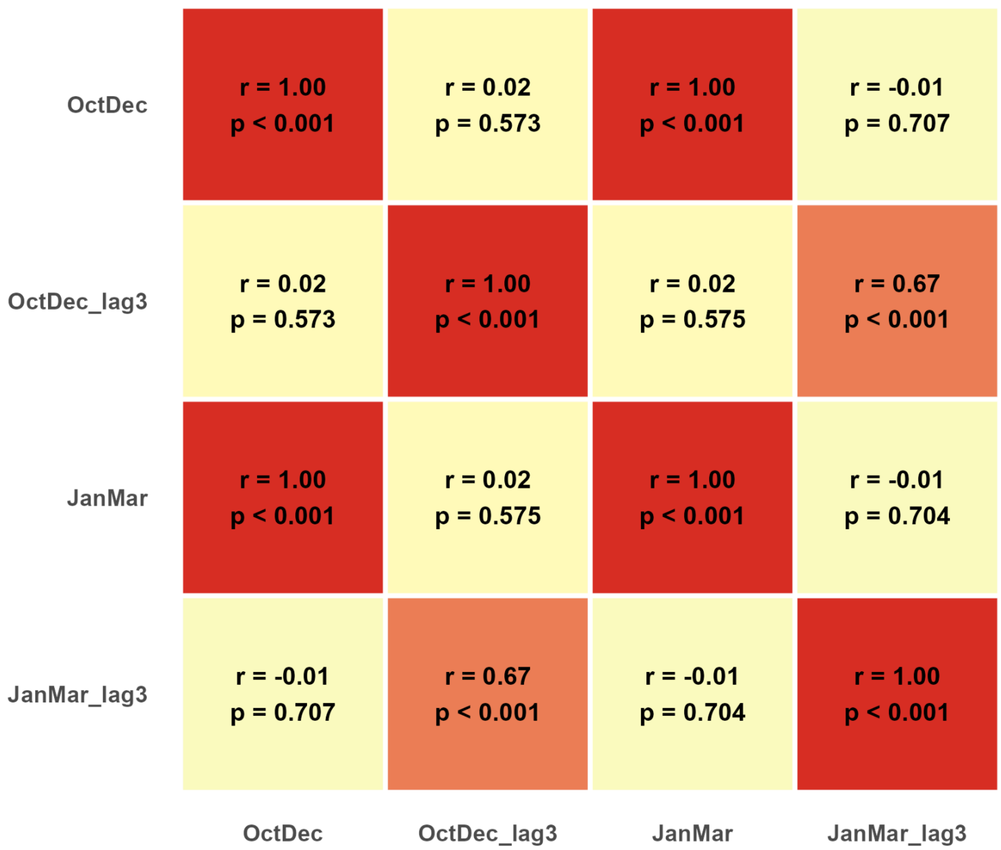
Assess how correlated the time steps variables PDSI are to each other

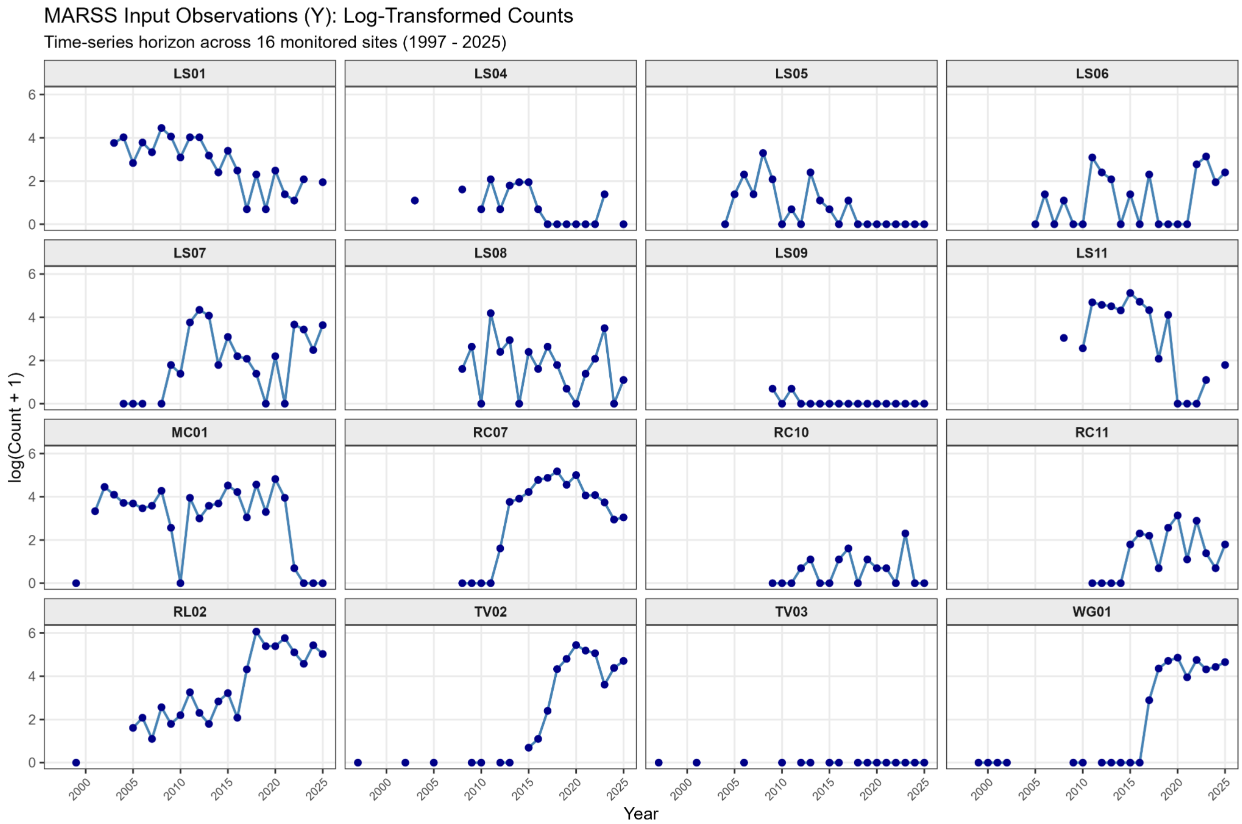
# Week 10

Aug 3, 2026

**PDSI Covariate Pearson Correlation Matrix**

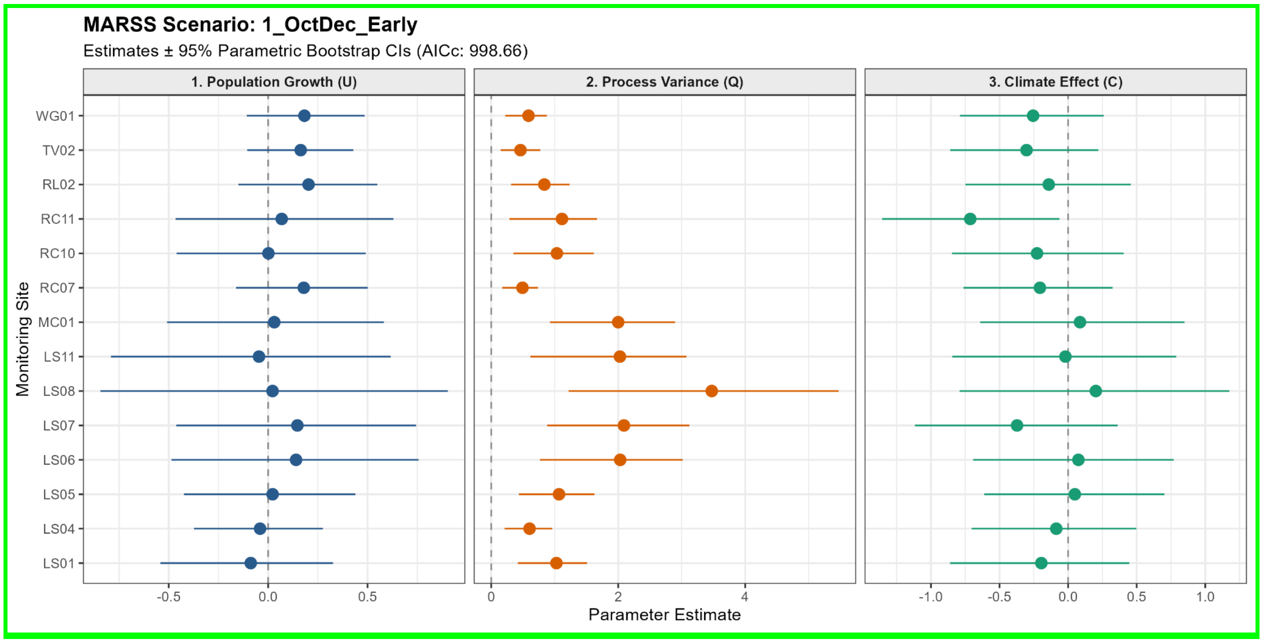
Current Seasons vs. T-3 Lags





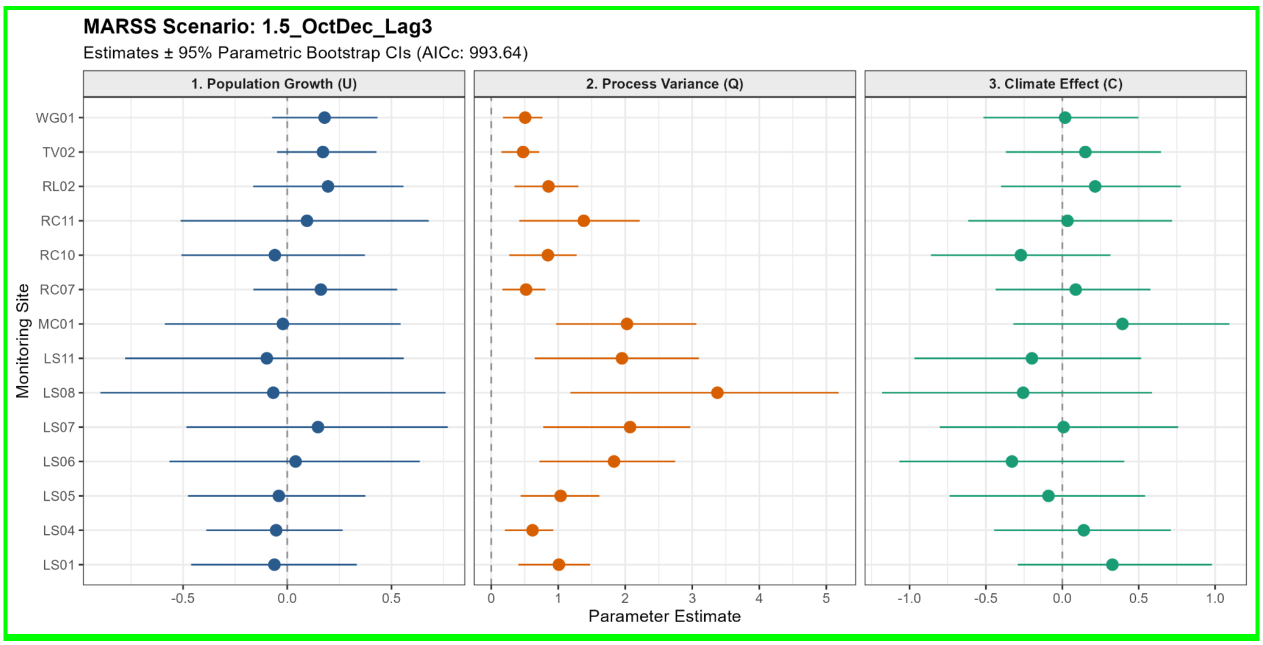

Z (connects observations to states) -> “identity”
R = “zero” (so it’d be MAR rather than MARSS)
B = “identity”
U = “unequal” (long-term trend)
Q = “diagonal and unequal” (allow each pond to have different levels of variability)
C = “Custom” Pond only responds to PDSI in its own watershed otherwise 0

New Model 1: Oct -Dec (Early) AIC: 998.66 4th lowest score

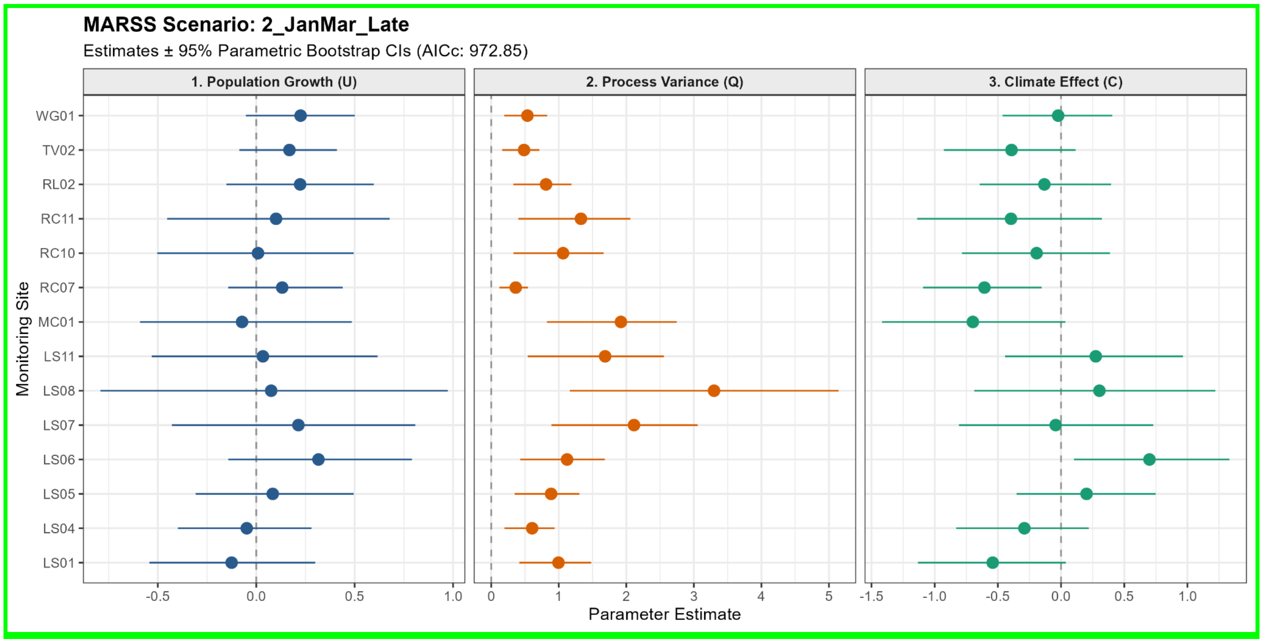



RC11: No overlap negative

New Model 1.5: Oct-Dec T-3 (Early lag) AIC: 993.64 third lowest score



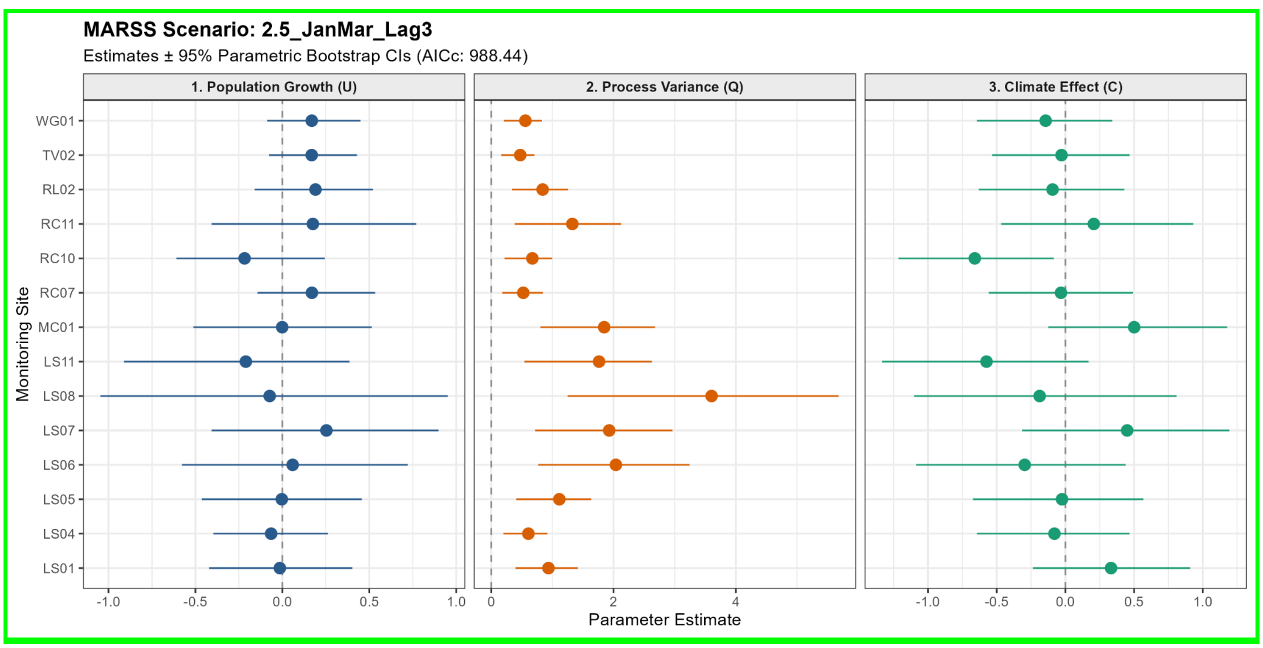

New Model 2: Jan-March (Late) AIC: 972.85 lowest score



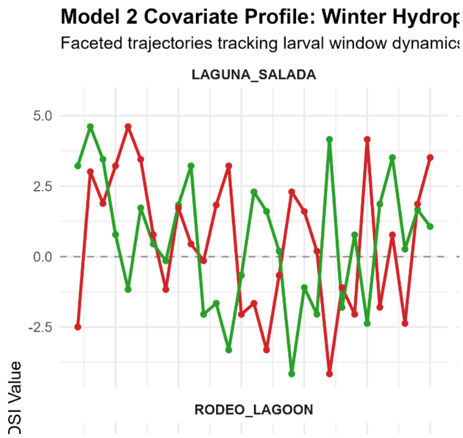

RCO7: No overlap negative

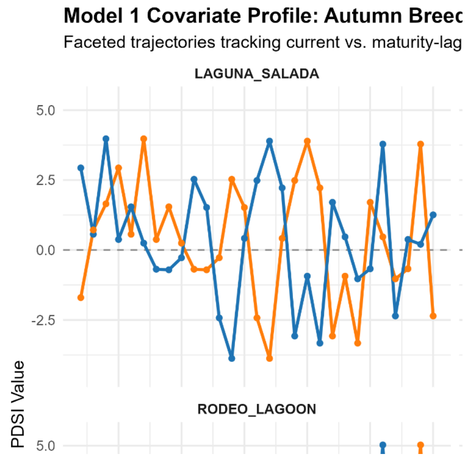
LS06: No overlap positive

New Model 2.5: Jan-March T-3 (Late lag) AIC: 988.44 second lowest score




RC10: No overlap




Watershed PDSI correlation overtime time series

1. Effect package plots r show response to pdsi

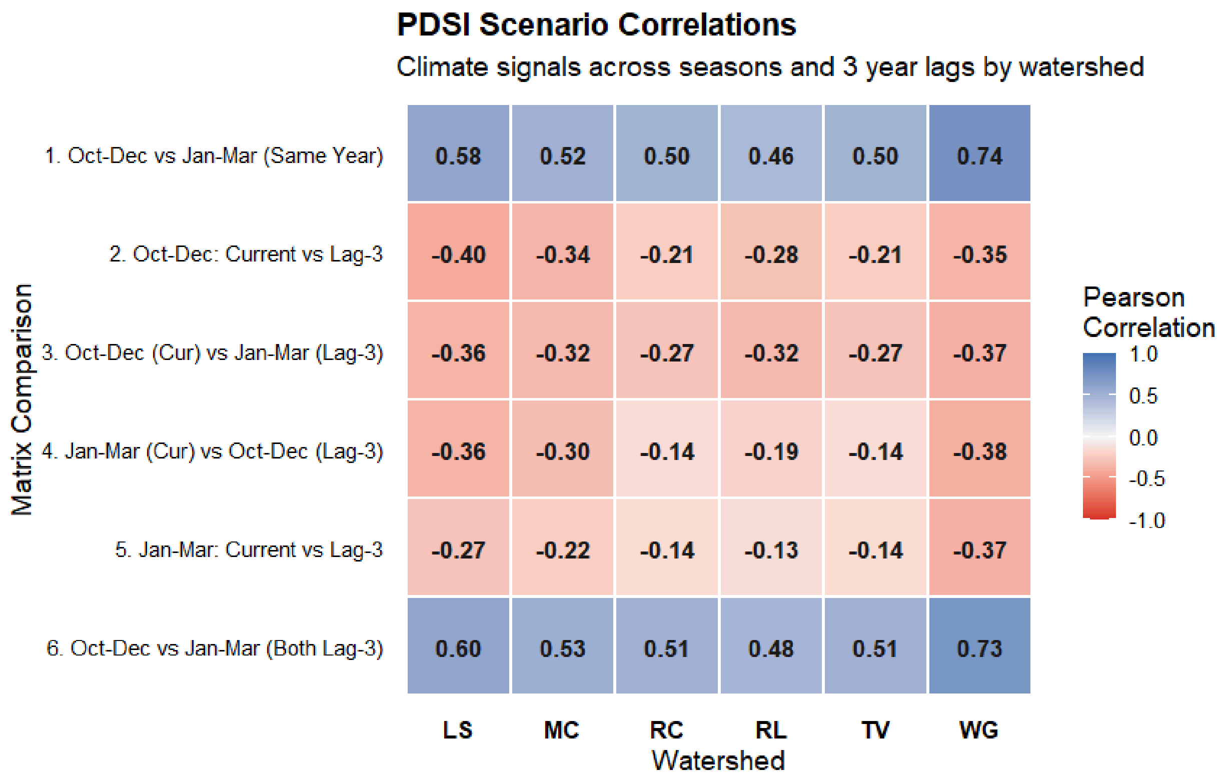
Additive model vs models with different interactions AIC

3 years ago it was wet so reproduction moved forward 3 years and it's wet. Those sexually mature females should reproduce. Should be multiplying instead of additive model we were running compare AIC rankings

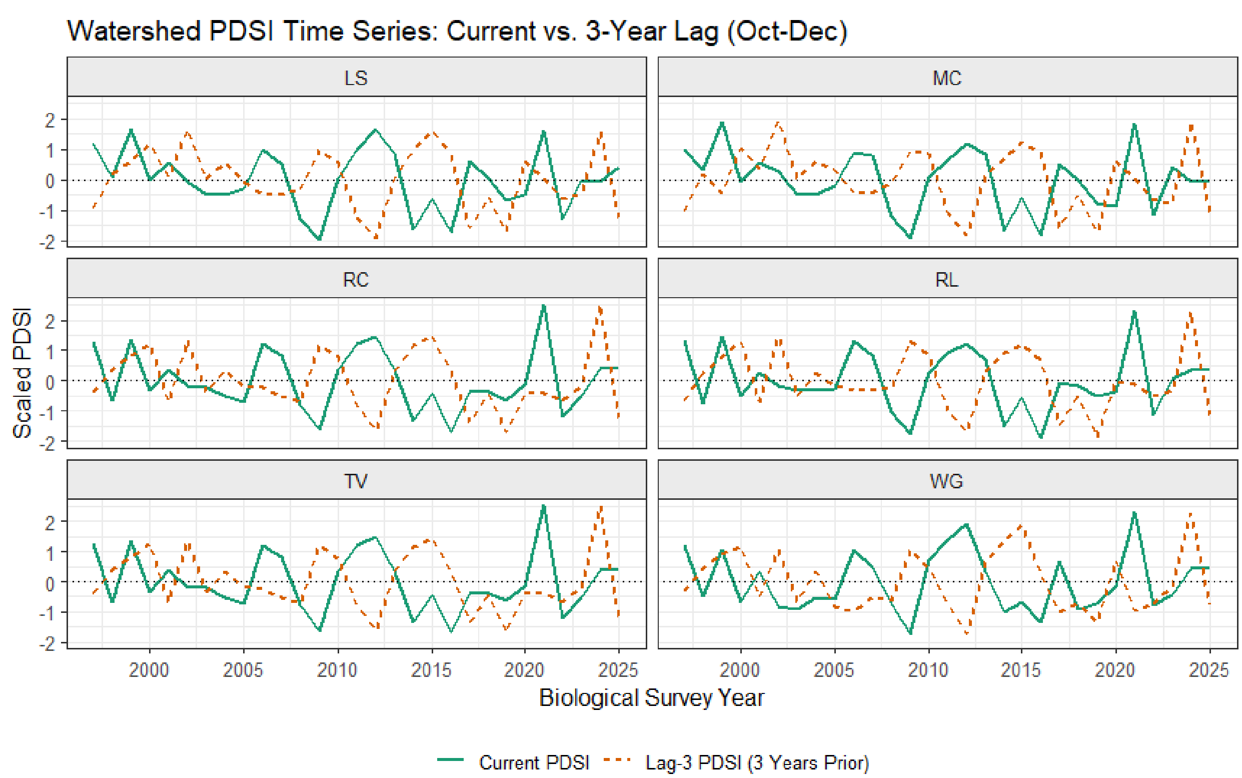
To do list

1. Time series for each individual watershed PDSI and T-3
2. Check the pdsi correlation heatmap
    a. Send over the R MARSS code
3. Look into "3 years ago it was wet so reproduction moved forward 3 years and it's wet. Those sexually mature females should reproduce. Should be multiplying instead of additive model we were running compare AIC rankings"
4. Model 1: Contemporary Early and late (Oct-Dec and Jan-March)
5. Model 2: Average between current and previous
6. Model 3: Average between current and minus 1 year and minus 2 year

7. PDSI CURRENT AND T-3 FOR EACH FOR WATERSHED




8.



9. Interactive models 9 models

- Early X late PDSI Oct-Dec Jan-March
    - Need them I
    - EggCount ~ PDSI(oct-dec) * PDSI(Jan-march)
        - Beta1 * PDSIoct-dec → ns
        - Beta2 * PDSIjan-march → ns
        - Beta3* PDSIoct-dec*PDSIjan-march → significant
- Early only PDSI Oct-Dec
    - Have it
- Late only PDSI Jan-March
    - Have it
- 3 As above with lag t-3 (all six reasonably)
    - Early T-3, late T-3 have it
- PDSI mean (0), (0+ t-1), (0 + t-1, t-2) + add the lag (t-3) covariates
- Model 7: Current Year + T-3
- Model 8: Current Year + T-1 + T-3
- Model 9: Current Year 3 year average + T-3
    - B1*mean(2025,2024,2023) + B2*2022
- Look at the residuals as well as the AIC scores
- Find function that plots the residuals best aic have good residuals

I need

1. Interactive Early x Late
    a. By itself and its T-3 Variant

2. Early season Rolling average and its T-3 variant
3. Make a table with the hypothesis behind each model covariates, egg counts, and AIC values, candidate models
4. LS 05 go back and check if it is at risk site
5. Dynamic cost surface modeling continue to make rescue matrix for B matrix
6. Push: Base Egg count, PDSI, Marss model code
7. PDSI correlation mark with asterisk for significance
8. Have MARSS: Code ready to share on Monday

To think about:

- Effects plots
- Table for models and hypotheses and AIC scores
- Residual plots of best model
- BB send example code for making effects plots from MAR coefficients and data

  Here is a toy example without errors:

  ```
  # Extract estimated covariate coefficient where fit = your marss model
  object

  C_est <- coef(fit, type = "matrix")$C

  # Build a range of covariate values
  cov_range <- seq(min(PDSI), max(PDSI), length.out = 100)

  # Effect on the process = C * covariate
  effect <- C_est[1,1] * cov_range

  plot(cov_range, effect, type = "l")
  ```

- Github push/pull
-

# Week 10.5

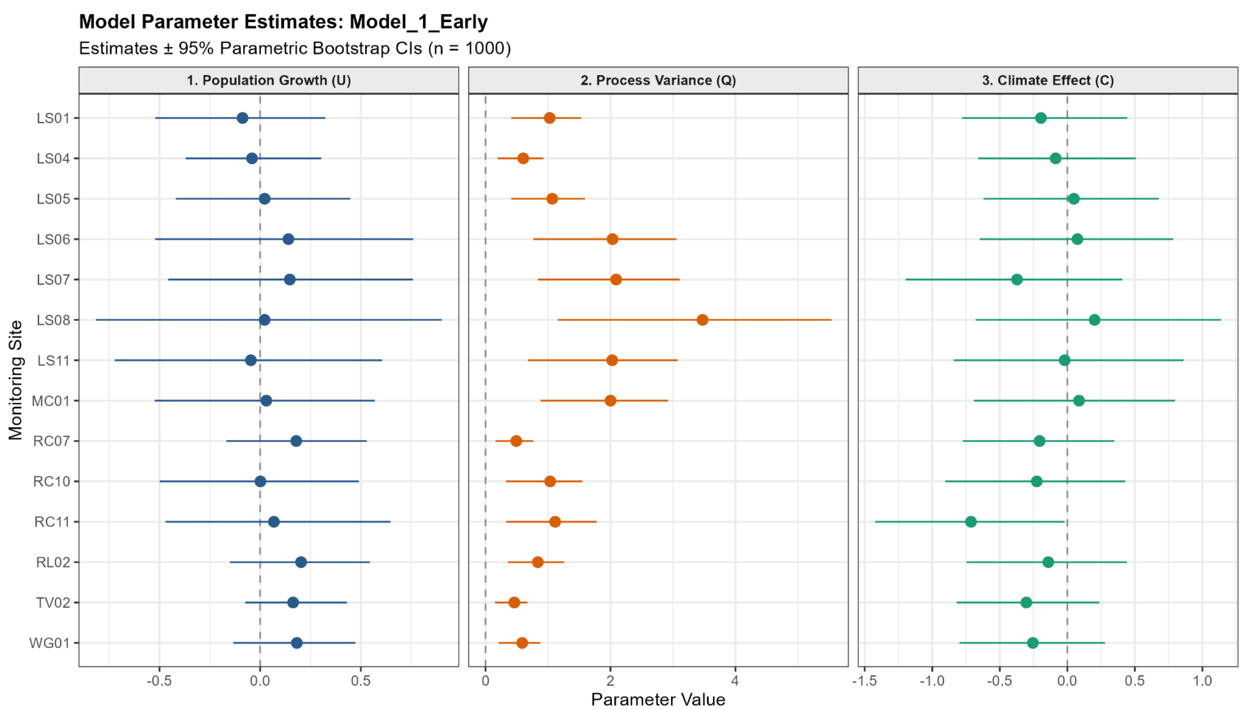
U. Trend of Population

Q: Random Biological Variance

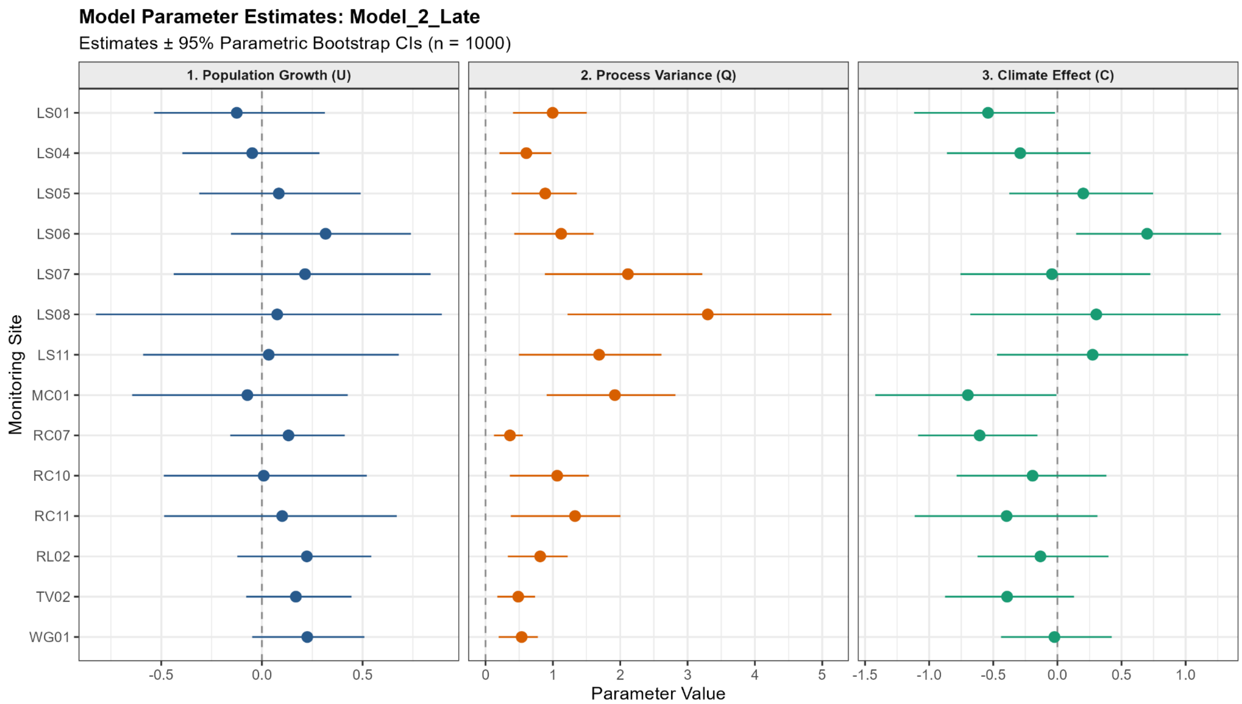
C: PDSI effect on population Sensitivity to climate

1. C x PDSI Score
   a. -0.60 x (+2.00 Wet) = -1.20 negative effect on population
   b. -0.60 x (-2.00 Dry) = 1.20 positive effect on population

Model 1: Early Oct-Dec

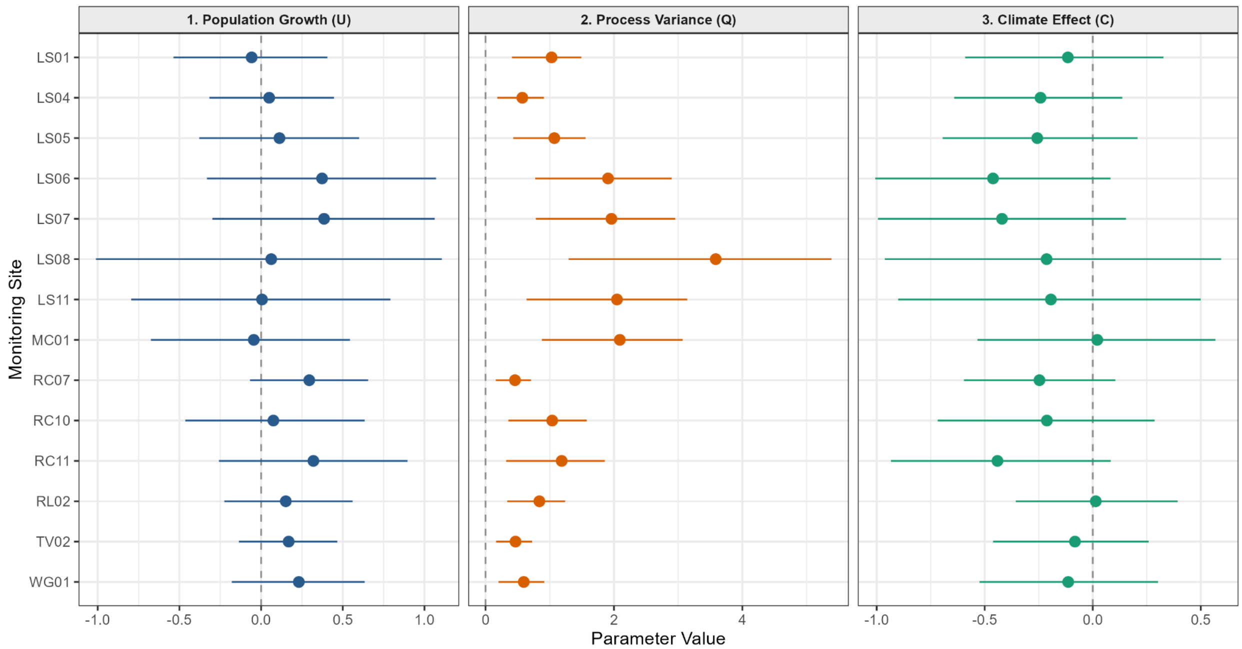


Model 2: Late Jan-March



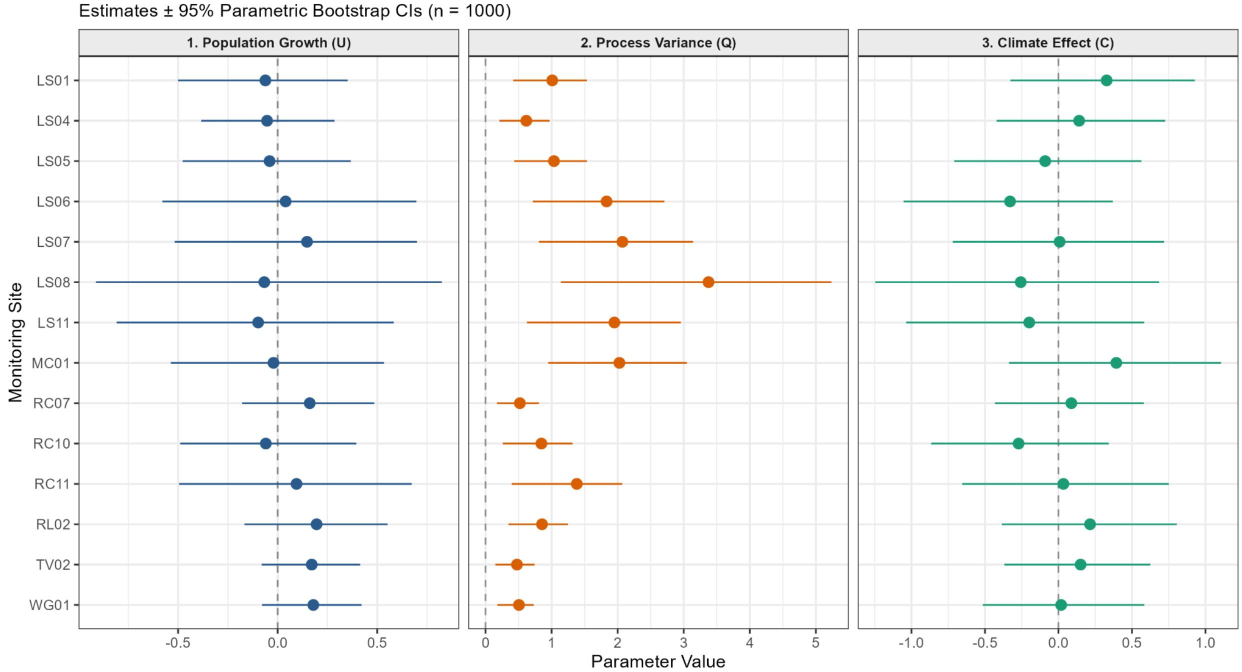

Model 3: Early X Late

**Model Parameter Estimates: Model_3_Inter_Cur**
Estimates ± 95% Parametric Bootstrap CIs (n = 1000)

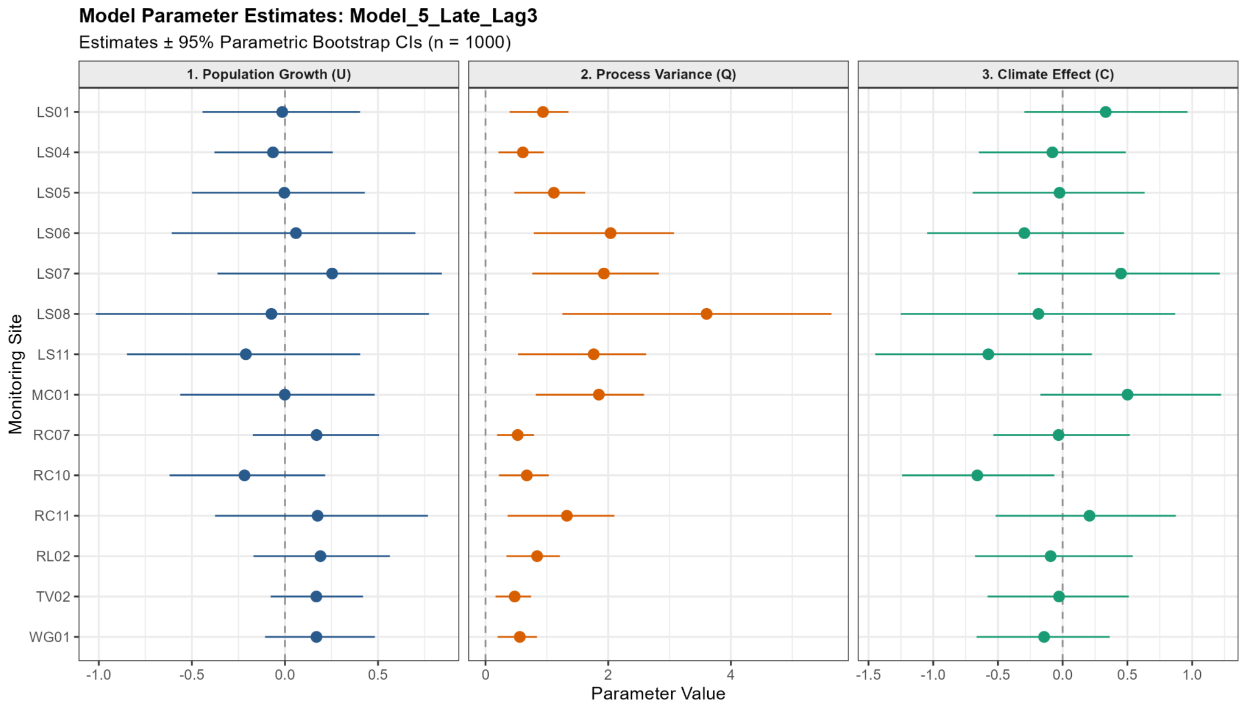



Model 4: Early Oct-Dec T-3 Only

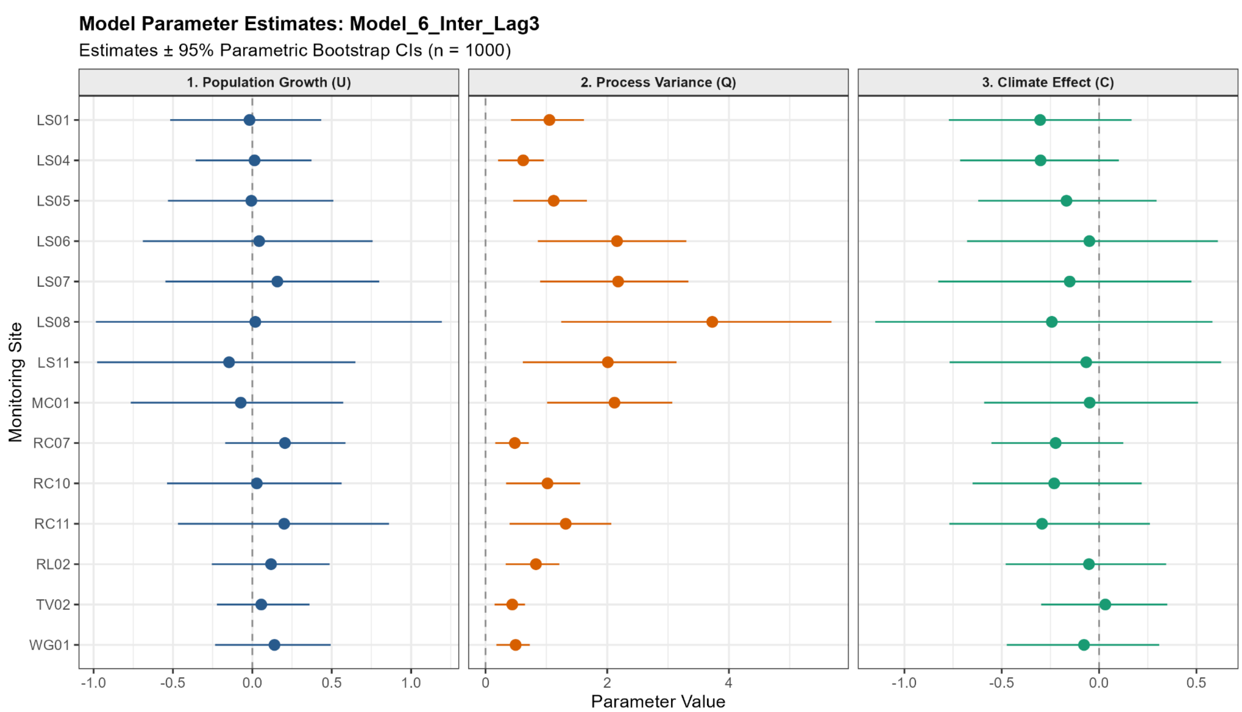
**Model Parameter Estimates: Model_4_Early_Lag3**
Estimates ± 95% Parametric Bootstrap CIs (n = 1000)



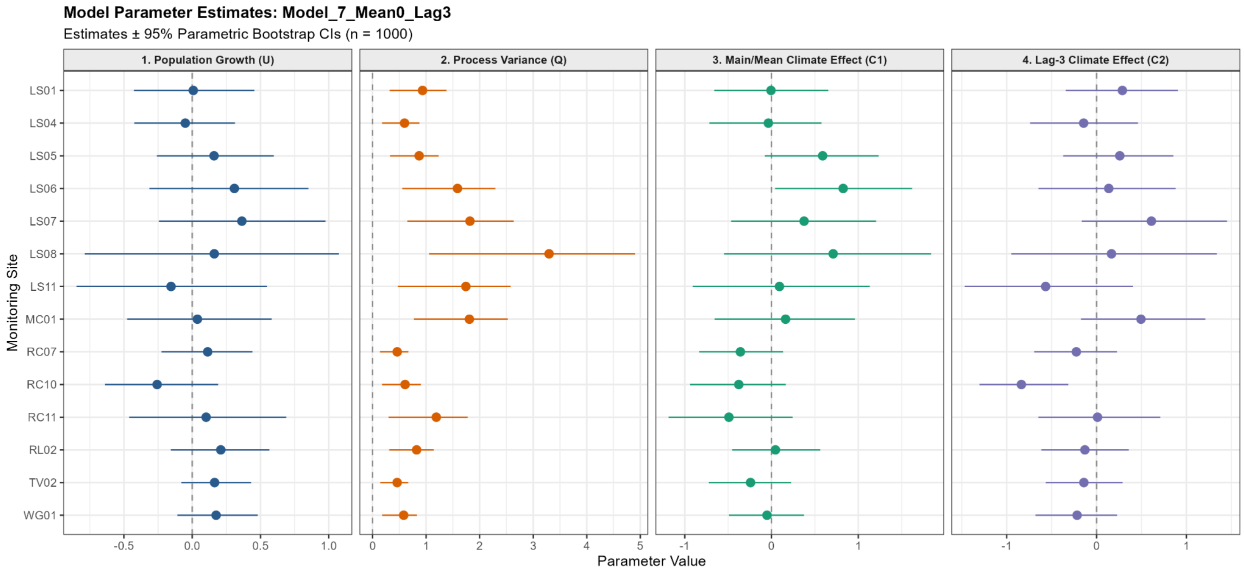

Model 5: Late Jan-March T-3 Only



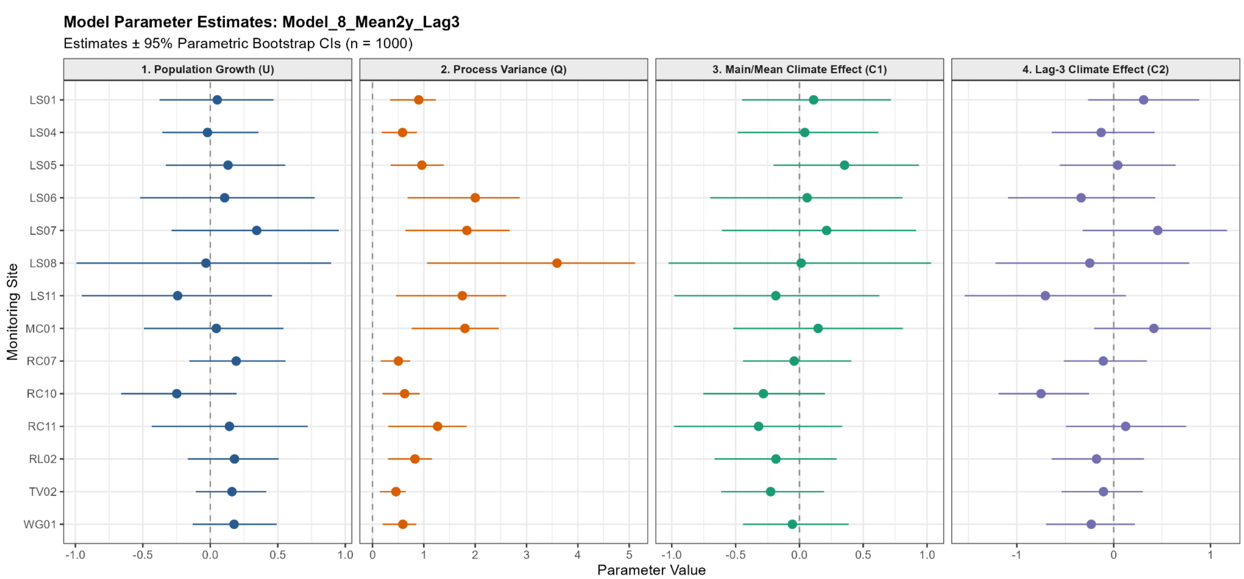

Model 6: Early X Late T-3 Only



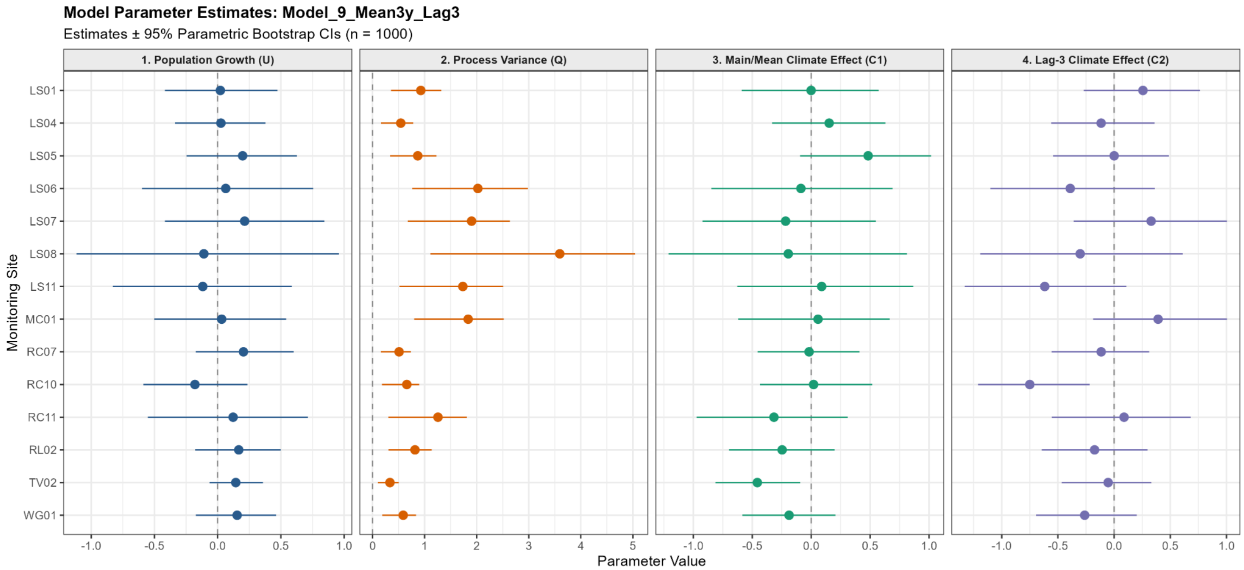

Model 7: Jan-March T-3




Model 8: Running 2 year mean + T-3



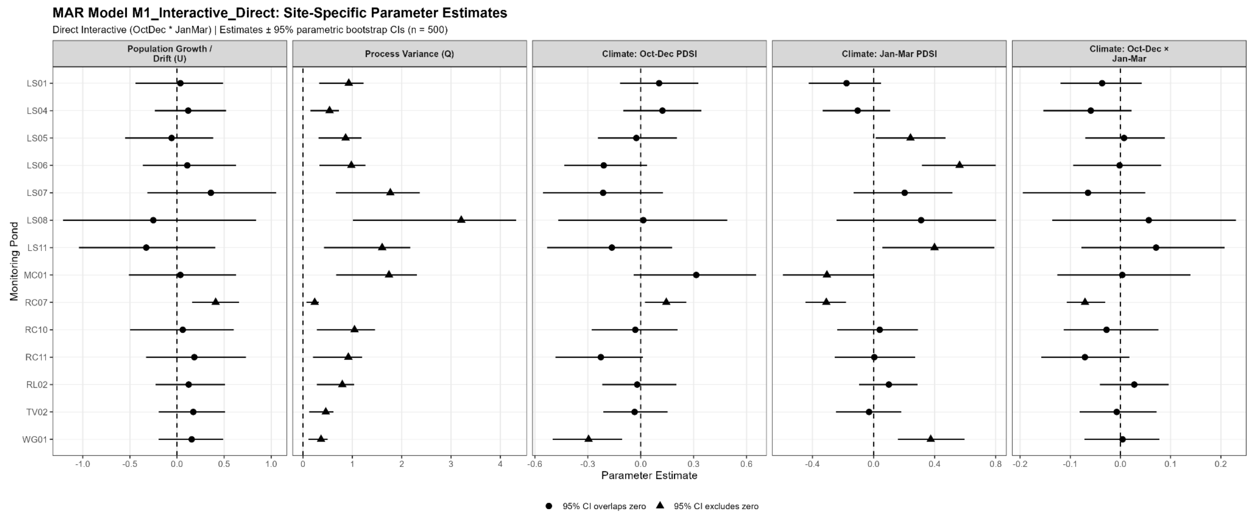

Model 9: Mean 3yr + T-3



Model Parameter Estimates: Model_9_Mean3y_Lag3
Estimates ± 95% Parametric Bootstrap CIs (n = 1000)
1. Population Growth (U)
2. Process Variance (Q)
3. Main/Mean Climate Effect (C1)
4. Lag-3 Climate Effect (C2)
Monitoring Site
LS01
LS04
LS05
LS06
LS07
LS08
LS11
MC01
RC07
RC10
RC11
RL02
TV02
WG01
-1.0
-0.5
0.0
0.5
1.0
0
1
2
3
4
5
Parameter Value

# Week 11

Aug 10, 2026

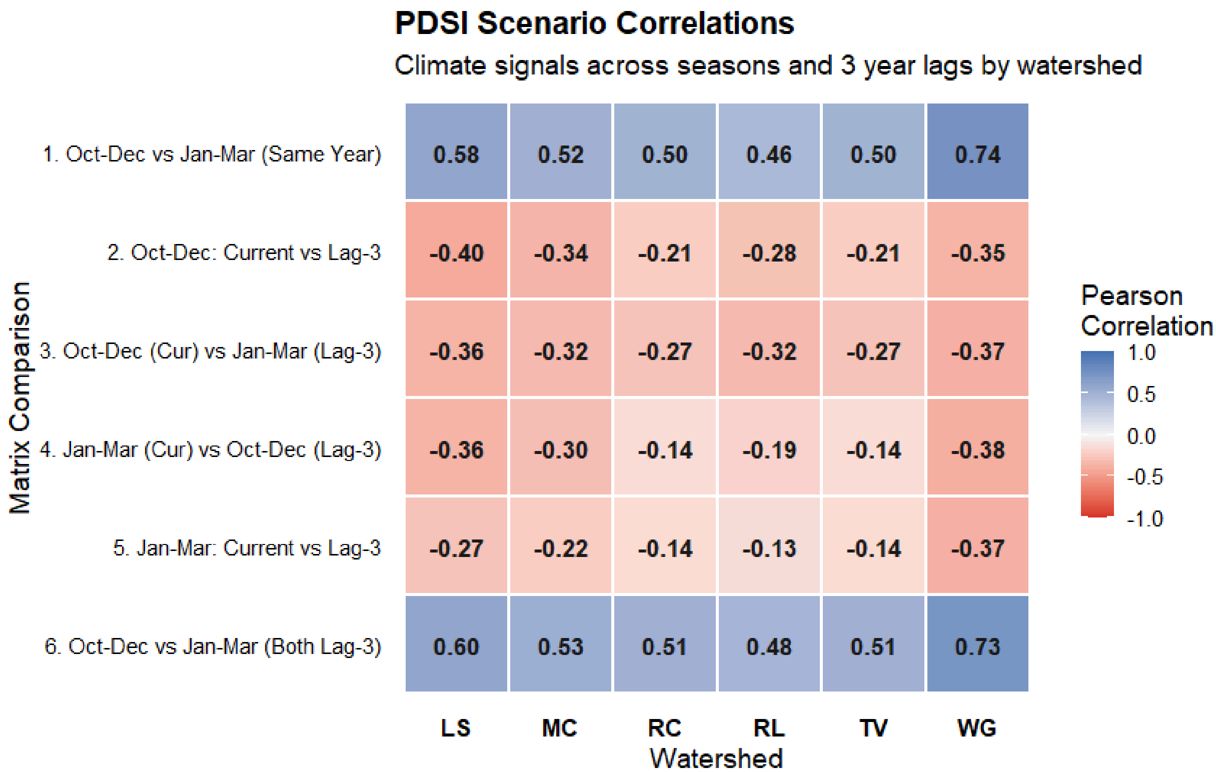



1.

Drop the 2 Covariates for Early and Late just keep the interactive Climate

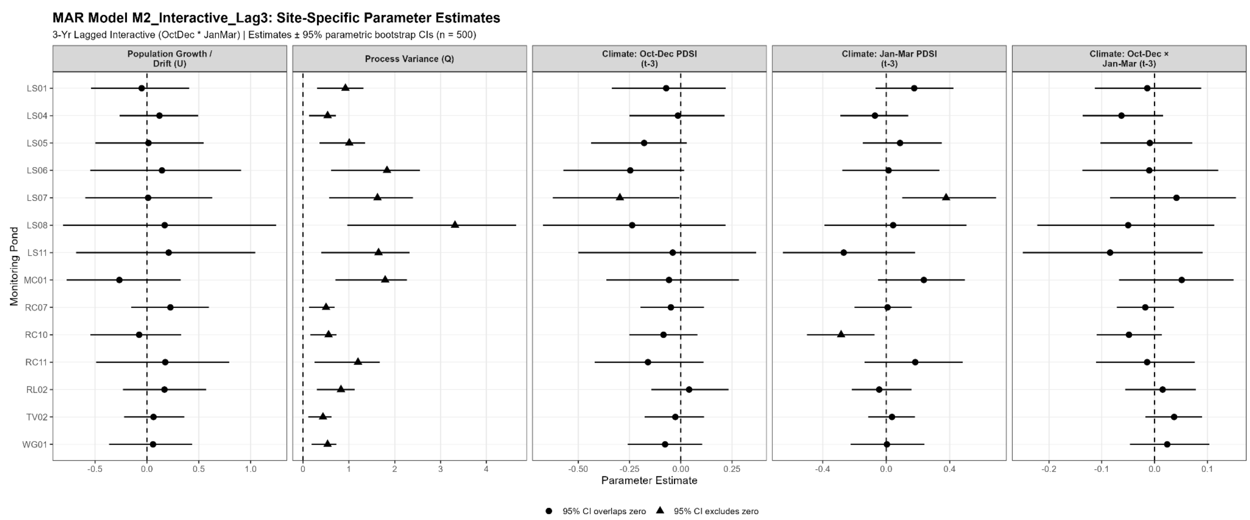
Covariate Result

Early x Late = interactive covariate + Early T-3 x Late T-3 = interactive covariate T-3



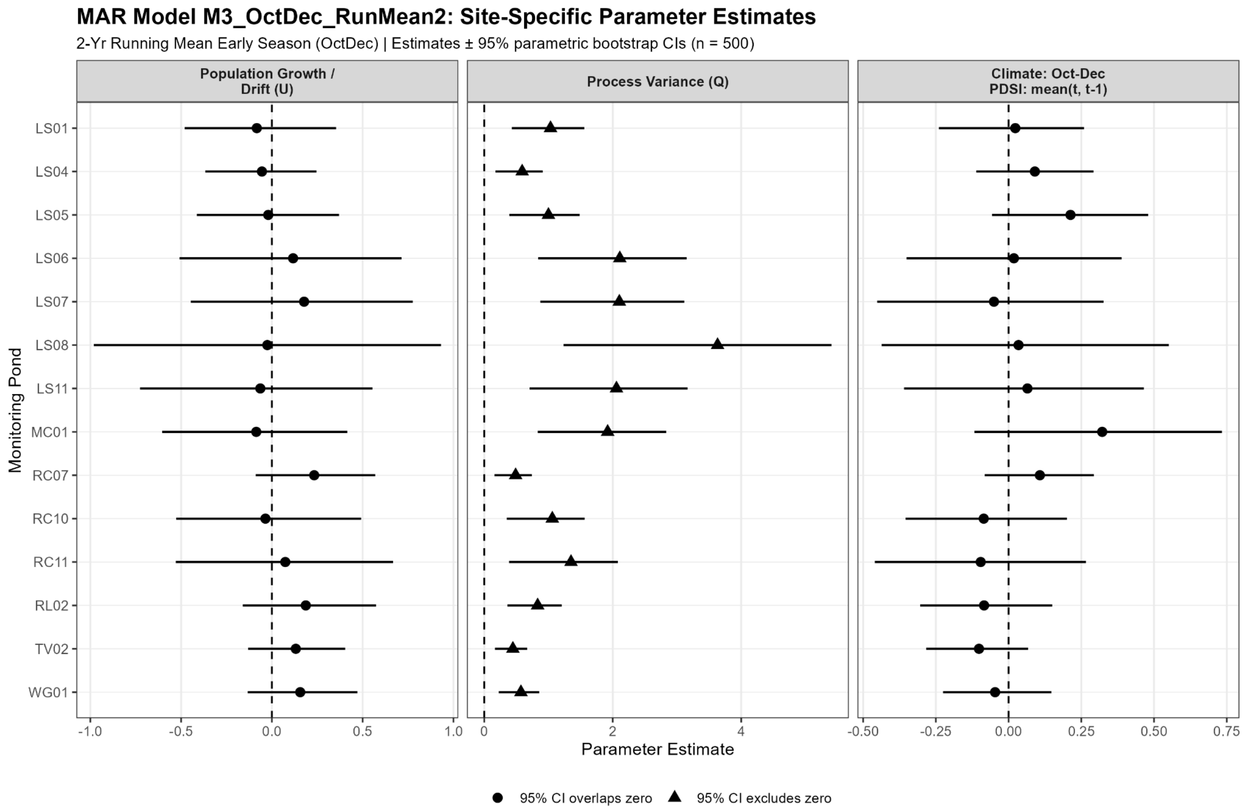

2.

RAW AIC: 958.52 AICc: 1034.48




3.

RAW AIC: 989.43 AICc: 1065.39



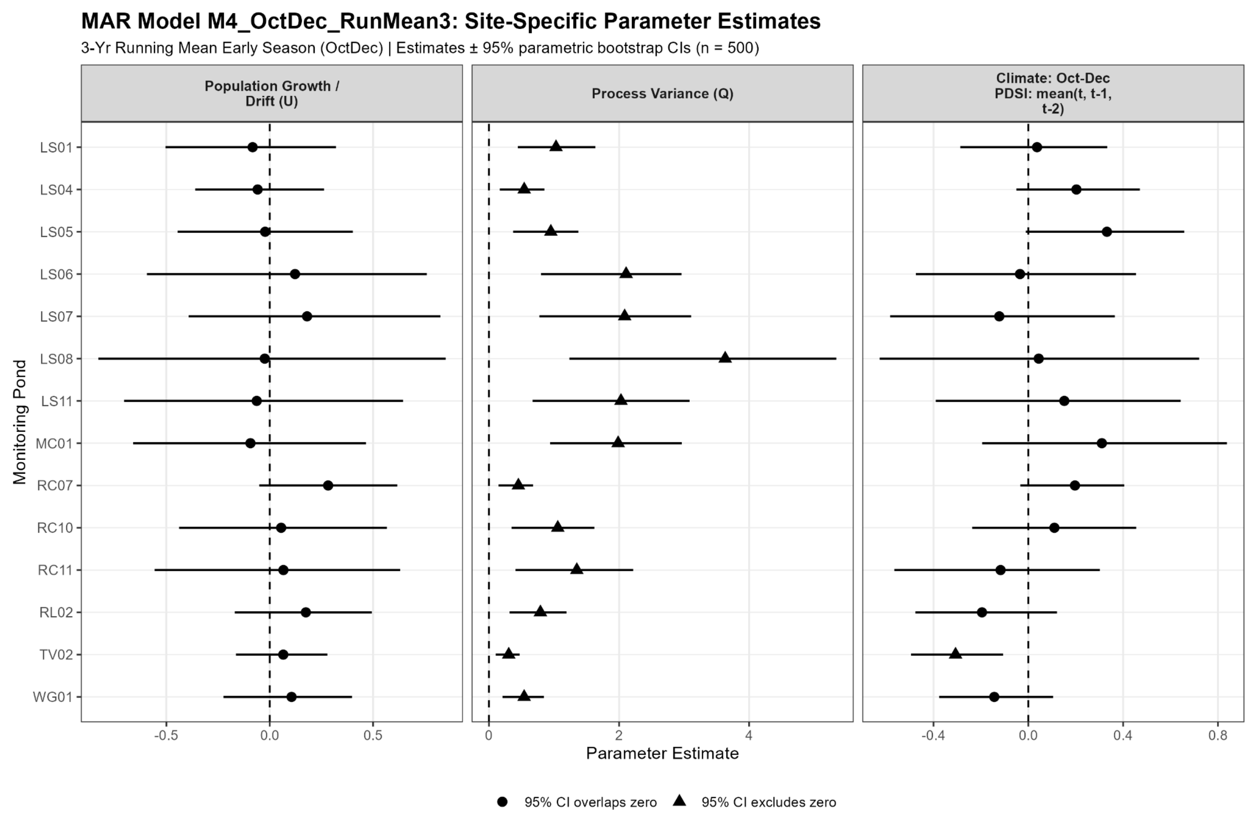

4.

RAW AIC: 966.69 AICc: 996.24

Average for Oct-dec (2025+2024) ( C +T-1) (2025+2024+2023) (C + T-1 + T-2)  T-3

2022

Hypothesis Covariate Model AIC Residuals




5.

RAW AIC: 954.20 AICc: 983.76

Hand off Requirements

Front to end Script

Table: Model #, Covariate, AIC, AICc, CI

Model: Early Oct-Dec Contemporary

Is early season (Oct-Dec) PDSI more important for the population in these individual sites?

Model: Late Jan-March Contemporary

Is late season (Jan-March) PDSI more important for the population in these individual sites?

Model: Early Oct-Dec Contemporary + T-3 Year lag

Does early season PDSI from 3 year prior (to give time for sexual maturity) have an effect on the population for concurrent year while accounting for same year climate conditions?

Model: Late Jan-March Contemporary + T-3 year lag

Model: Early X Late Contemporary

Do both early and late season have effects on population growth?

Model: Early Running 3 year mean

(2025,2024,2023)

Does the running mean for 3 years of climate affect the egg mass abundance?

Model: Late Running 3 year mean

(2025,2024,2023)

Model: Early Oct-Dec Recent (3 year) mean + T-3 Year lag (2022)

(2025,2024,2023) + (2022)

Early climate conditions over the last 3 years + 3 years prior affect egg mass abundance?

(T0,T-1,T-2) + (T-3)

Model: Late Running 3 year mean + T-3 Year lag

Model: Just egg mass abundance

Plot the coefficients for the best supportive model

Next steps:

Readme.txt for all the important variables

Thinking behind the decisions what and why

Egg mass counts 1997-2025

Lat and Lon Coordinates

Site Notes

PDSI WWDT link

Marss Model Script

C-Matrix breakdown

Methods for the data handling

Steps I took to move it from raw to end file

Treat it like usermanual